

THE TT-GEOMETRY OF PERMUTATION MODULES. PART II: TWISTED COHOMOLOGY

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ABSTRACT. We continue our analysis of the derived category of permutation modules over a finite group. In Part I, we identified the spectrum of this tensor-triangulated category as a set. Here we describe its topology, by reducing the problem to elementary abelian groups and then using a twisted form of cohomology to realize the spectrum as a Dirac scheme.

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1. INTRODUCTION

Unless mentioned otherwise, G stands for a finite group and k for a field of positive characteristic p , with p typically dividing the order of G .

Executive summary. We study the homotopy category of bounded complexes of permutation kG -modules, idempotent-completed:

$$\mathcal{K}(G) := K_b(\text{perm}(G; k))^{\natural} = K_b(\text{perm}(G; k)^{\natural}).$$

This $\mathcal{K}(G)$ is a tensor-triangulated category (‘tt-category’ for short). We determined all the points of its tt-spectrum $\text{Spc}(\mathcal{K}(G))$ in Part I of this series; see [BG22]. The present paper aims at elucidating its *topology*. This knowledge will give us, among other things, the classification of thick $\otimes$ -ideals in $\mathcal{K}(G)$.

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Our main results are the following. On the one hand, we show that the space $\mathrm{Spc}(\mathcal{K}(G))$ is a colimit of $\mathrm{Spc}(\mathcal{K}(E))$ over a suitable category of elementary abelian p -groups E that appear as subquotients of G . On the other hand, when E is elementary abelian, we describe the spectrum $\mathrm{Spc}(\mathcal{K}(E))$ as a ‘Dirac scheme’ in the sense of Hesselholt-Pstrągowski [HP23]. Combining these results yields a description of the topological space $\mathrm{Spc}(\mathcal{K}(G))$ for all G . Let us now explain these ideas.

The colimit theorem. To discuss the tt-geometry of $\mathcal{K}(G)$, it is instructive to keep in mind the bounded derived category of finitely generated kG -modules, $D_b(kG)$, which is a localization of our $\mathcal{K}(G)$ by [BG23, Theorem 5.13]. A theorem of Serre [Ser65], famously expanded by Quillen [Qui71], implies that $\mathrm{Spc}(D_b(kG))$ is the colimit of the $\mathrm{Spc}(D_b(kE))$, for E running through the elementary abelian p -subgroups of G ; see [Bal16, §4]. The indexing category for this colimit is an orbit category: Its morphisms keep track of conjugations and inclusions of subgroups.

In Part I, we proved that $\mathrm{Spc}(\mathcal{K}(G))$ is set-theoretically partitioned into spectra of derived categories $D_b(k(G//K))$ for certain subquotients of G , namely the Weyl groups $G//K = (N_G K)/K$ of p -subgroups $K \leq G$. It is then natural to expect a more intricate analogue of Quillen’s result for the tt-category $\mathcal{K}(G)$, in which subgroups are replaced by *subquotients*. This is precisely what we prove. The orbit category has to be replaced by a category $\mathcal{E}_p(G)$ whose objects are elementary abelian p -sections $E = H/K$, for p -subgroups $K \trianglelefteq H \leq G$. The morphisms in $\mathcal{E}_p(G)$ keep track of conjugations, inclusions and *quotients*. See Construction 2.1.

This allows us to formulate our reduction to elementary abelian groups:

1.1. **Theorem** (Theorem 2.10). *There is a canonical homeomorphism*

$$\mathrm{colim}_{E \in \mathcal{E}_p(G)} \mathrm{Spc}(\mathcal{K}(E)) \xrightarrow{\sim} \mathrm{Spc}(\mathcal{K}(G)).$$

The category $\mathcal{E}_p(G)$ has been considered before, e.g. in Bouc-Thévenaz [BT08]. Every morphism in $\mathcal{E}_p(G)$ is the composite of three special morphisms (Remark 2.3)

$$(1.2) \quad E \xrightarrow{!} E' \rightarrow E'' \xrightarrow{\sim} E'''$$

where $E = E'/N$ is a quotient of E' (sic!), where $E' \leq E''$ is a subgroup of E'' and where E''' is a G -conjugate of E'' . The tt-category $\mathcal{K}(E)$ is contravariant in $E \in \mathcal{E}_p(G)$ and the tt-functors corresponding to (1.2)

$$(1.3) \quad \mathcal{K}(E''') \xrightarrow{\sim} \mathcal{K}(E'') \xrightarrow{\mathrm{Res}} \mathcal{K}(E') \xrightarrow{\Psi^N} \mathcal{K}(E)$$

yield the standard conjugation isomorphism, the standard restriction functor, and the less standard *modular N -fixed-points functor* Ψ^N introduced in Part I. The latter is a type of Brauer quotient that makes sense on the homotopy category of permutation modules. Such functors Ψ^N do not exist on derived or stable categories and they distinguish our results and their proofs from the classical theory. It is an open question whether they will also play a role in the generality of [BCH⁺23].

Twisted cohomology. The above discussion reduces the analysis of $\mathrm{Spc}(\mathcal{K}(G))$ to the case of elementary abelian p -groups E . As often in modular representation theory, this case is far from trivial and can be viewed as the heart of the matter.

So let E be an elementary abelian p -group. Our methods will rely on $\otimes$ -invertible objects u_N in $\mathcal{K}(E)$ indexed by the set $\mathcal{N}(E) = \{ N \triangleleft E \mid [E:N] = p \}$ of maximal subgroups. These objects are of the form $u_N = (0 \rightarrow k(E/N) \rightarrow k(E/N) \rightarrow k \rightarrow 0)$

for p odd and $u_N = (0 \rightarrow k(E/N) \rightarrow k \rightarrow 0)$ for $p = 2$. See Definition 3.3. We use these $\otimes$ -invertibles u_N to construct a multi-graded ring

$$(1.4) \quad H^{\bullet\bullet}(E) = \bigoplus_{s \in \mathbb{Z}} \bigoplus_{q \in \mathbb{N}^{N(E)}} \mathrm{Hom}_{\mathcal{K}(E)}(\mathbb{1}, \mathbb{1}(q)[s]),$$

where $\mathbb{1}(q)$ is the $\otimes$ -invertible $\bigotimes_{N \in N(E)} u_N^{\otimes q(N)}$ for every tuple $q: N(E) \rightarrow \mathbb{N}$, that we refer to as a ‘twist’. Without these twists we would obtain the standard $\mathbb{Z}$ -graded endomorphism ring $\mathrm{End}^\bullet(\mathbb{1}) := \bigoplus_{s \in \mathbb{Z}} \mathrm{Hom}(\mathbb{1}, \mathbb{1}[s])$ of $\mathbb{1}$ which, for $D_b(kE)$, is identified with the cohomology $H^\bullet(E, k)$, but for $\mathcal{K}(E)$ is reduced to the field k and therefore rather uninteresting. We call $H^{\bullet\bullet}(E)$ the *(permutation) twisted cohomology* of E . Some readers may appreciate the analogy with cohomology twisted by line bundles in algebraic geometry, or with Tate twists in motivic cohomology.

We can employ this multi-graded ring $H^{\bullet\bullet}(E)$ to describe $\mathrm{Spc}(\mathcal{K}(E))$:

1.5. Theorem (Corollary 6.6). *The space $\mathrm{Spc}(\mathcal{K}(E))$ identifies with an open subspace of the homogeneous spectrum of $H^{\bullet\bullet}(E)$ via a canonical ‘comparison map’.*

The comparison map in question generalizes the one of [Bal10], which landed in the homogenous spectrum of $\mathrm{End}^\bullet(\mathbb{1})$ without twist. We also describe in Corollary 6.6 the open image of this map by explicit equations in $H^{\bullet\bullet}(E)$.

Dirac geometry. If the reader is puzzled by the multi-graded ring $H^{\bullet\bullet}(E)$, here is another approach based on a special open cover $\{U(H)\}_{H \leq E}$ of $\mathrm{Spc}(\mathcal{K}(E))$ indexed by the subgroups of E and introduced in Proposition 4.11. Its key property is that over each open $U(H)$ all the $\otimes$ -invertible objects u_N are trivial: $(u_N)_{|U(H)} \simeq \mathbb{1}[s]$ for some shift $s \in \mathbb{Z}$ depending on H and N . For the trivial subgroup $H = 1$, the open $U(1)$ is the ‘cohomological open’ of Part I, that corresponds to the image under $\mathrm{Spc}(-)$ of the localization $\mathcal{K}(E) \rightarrow D_b(kE)$. See Proposition 4.14. At the other end, for $H = E$, we show in Proposition 4.17 that the open $U(E)$ is the ‘geometric open’ that corresponds to the localization of $\mathcal{K}(E)$ given by the geometric fixed-points functor. Compare [BG22, Remark 3.11]. For E cyclic, these two opens $U(1)$ and $U(E)$ are all there is to consider. But as the p -rank of E grows, there is an exponentially larger collection $\{U(H)\}_{H \leq E}$ of open subsets interpolating between $U(1)$ and $U(E)$. This cover $\{U(H)\}_{H \leq E}$ allows us to use the classical comparison map of [Bal10] locally. It yields a homeomorphism between each $U(H)$ and the homogeneous spectrum of the $\mathbb{Z}$ -graded endomorphism ring $\mathrm{End}_{U(H)}^\bullet(\mathbb{1})$ in the localization $\mathcal{K}(E)_{|U(H)}$. In compact form, this can be rephrased as follows (a Dirac scheme is to a usual scheme what a $\mathbb{Z}$ -graded ring is to a non-graded one):

1.6. Theorem (Corollary 6.4). *The space $\mathrm{Spc}(\mathcal{K}(E))$, together with the sheaf of $\mathbb{Z}$ -graded rings obtained locally from endomorphisms of the unit, is a Dirac scheme.*

Elementary abelian take-home. Let us ponder the $\mathbb{Z}$ -graded endomorphism ring of the unit $\mathrm{End}^\bullet(\mathbb{1})$ for a moment. Recall from [BCR97, BIK11] that the spectrum of the derived category, $\mathrm{Spc}(D_b(kE))$, is homeomorphic to the homogeneous spectrum of the cohomology ring $H^\bullet(E, k) \cong \mathrm{End}_{D_b(kE)}^\bullet(\mathbb{1})$, see [Bal10, Proposition 8.5]. Such a result cannot hold for $\mathcal{K}(E)$ since the ring $\mathrm{End}_{\mathcal{K}(E)}^\bullet(\mathbb{1}) = k$ is too small to provide geometric information. So we have developed two substitutes. Our first approach is to replace the usual $\mathbb{Z}$ -graded ring $\mathrm{End}^\bullet(\mathbb{1})$ by a richer multi-graded ring involving twists. This leads us to twisted cohomology $H^{\bullet\bullet}(E)$

and to Theorem 1.5. The second approach is to hope that the endomorphism ring $\text{End}^\bullet(\mathbb{1})$, although useless globally, becomes rich enough to control the topology *locally* on $\text{Spc}(\mathcal{K}(E))$, without leaving the world of $\mathbb{Z}$ -graded rings. This is what we achieve in Theorem 1.6 thanks to the open cover $\{U(H)\}_{H \leq E}$. As can be expected, the two proofs are intertwined.

Touching ground. Combining Theorems 1.1 and 1.6 ultimately describes the topological space $\text{Spc}(\mathcal{K}(G))$ for all G , in terms of homogeneous spectra of graded rings. In Sections 7 to 9 we improve and apply these results as follows.

In Section 7, we explain how to go from the ‘local’ rings $\text{End}_{U(H)}^\bullet(\mathbb{1})$ over the open $U(H)$, for each subgroup $H \leq E$, to the ‘global’ topology of $\text{Spc}(\mathcal{K}(E))$.

In Theorem 8.13, we give a finite presentation by generators and relations of the reduced k -algebra $(\text{End}_{U(H)}^\bullet(\mathbb{1}))_{\text{red}}$, generalizing the usual one for cohomology.

In Corollary 9.12, we express $\text{Spc}(\mathcal{K}(G))$ for a general finite group G as the quotient of a disjoint union of $\text{Spc}(\mathcal{K}(E))$ for the *maximal* elementary abelian p -sections E of G by *maximal* relations.

In Proposition 9.14, we prove that the irreducible components of $\text{Spc}(\mathcal{K}(G))$ correspond to the maximal elementary abelian p -sections of G up to conjugation. It follows that the Krull dimension of $\text{Spc}(\mathcal{K}(G))$ is the sectional p -rank of G , the maximal rank of elementary abelian p -sections. (For comparison, recall that for the derived category these irreducible components correspond to maximal elementary abelian p -subgroups, not sections, and the Krull dimension is the usual p -rank.)

And of course, we discuss examples. Using our techniques, we compute $\text{Spc}(\mathcal{K}(G))$ for some notable groups G , in particular Klein-four (Example 7.16) and the dihedral group (Example 9.17). The latter will lead us to the following picture, whose precise meaning will be explained in Section 9. Hopefully, its beauty will entice the reader to proceed beyond the present introduction.

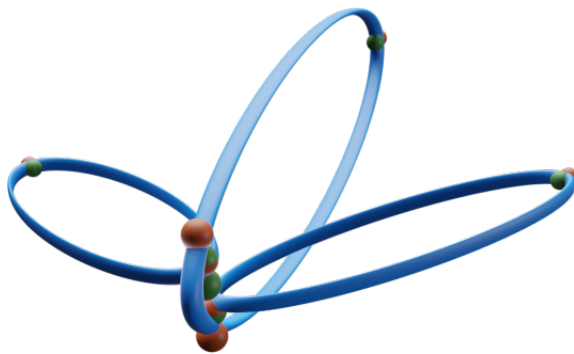


FIGURE 1. Three $\mathbb{P}_k^1$ with several points doubled and some identified.

The toolbox. The outline of the paper should now be clear from the above discussion and the table of content presented upfront. It would be an oversell to pretend that Part II is a stand-alone paper. We import several technical results from Part I, as black boxes. The ones invoked most often are gathered in Recollection 1.8.

Here is some standard notation used throughout the text.

1.7. *Notation.* We write $\mathrm{Spc}(\mathcal{K})$ for the tt-spectrum of a tt-category $\mathcal{K}$. For an object $x \in \mathcal{K}$, we write $\mathrm{open}(x) = \{ \mathcal{P} \in \mathrm{Spc}(\mathcal{K}) \mid x \in \mathcal{P} \}$ to denote the open complement of $\mathrm{supp}(x)$. We write $\mathrm{Sub}_p(G)$ for the set of p -subgroups of G and $\mathrm{Sub}_p(G)/_G$ for its G -orbits under G -conjugation. For each subgroup $H \leq G$, its Weyl group is $G//H = (N_G H)/H$ where $N_G H = \{ g \in G \mid H^g = H \}$.

Let us remind the reader of the essentials of Part I [BG22].

1.8. *Recollection.* The canonical localization $\Upsilon_G: \mathcal{K}(G) \rightarrow \mathrm{D}_b(kG)$ gives us an open piece $V_G := \mathrm{Spc}(\mathrm{D}_b(kG)) \cong \mathrm{Spec}^h(\mathrm{H}^\bullet(G, k))$ of the spectrum, that we call the ‘cohomological open’. We write $v_G = \mathrm{Spc}(\Upsilon_G): V_G \hookrightarrow \mathrm{Spc}(\mathcal{K}(G))$ for the inclusion. For every $H \in \mathrm{Sub}_p(G)$ we denote by $\Psi^H: \mathcal{K}(G) \rightarrow \mathcal{K}(G//H)$ the *modular H -fixed-points* tt-functor constructed in [BG22, Section 4]. It is characterized by $\Psi^H(k(X)) \simeq k(X^H)$ on permutation modules and by the same formula degreewise on complexes. We write $\check{\Psi}^H = \Upsilon_{G//H} \circ \Psi^H$ for the composite $\mathcal{K}(G) \rightarrow \mathcal{K}(G//H) \rightarrow \mathrm{D}_b(k(G//H))$ all the way down to the derived category of $G//H$. For every $H \in \mathrm{Sub}_p(G)$, the tt-prime $\mathcal{M}(H) = \mathrm{Ker}(\check{\Psi}^H)$ is a closed point of $\mathrm{Spc}(\mathcal{K}(G))$. It is also $\mathcal{M}(H) = \mathrm{Ker}(\mathbb{F}^H)$ where $\mathbb{F}^H = \mathrm{Res}_1^{G//H} \circ \Psi^H: \mathcal{K}(G) \rightarrow \mathcal{K}(G//H) \rightarrow \mathrm{D}_b(k)$. All closed points of $\mathrm{Spc}(\mathcal{K}(G))$ are of this form by [BG22, Corollary 6.31]. We write $\psi^H = \mathrm{Spc}(\Psi^H): \mathrm{Spc}(\mathcal{K}(G//H)) \rightarrow \mathrm{Spc}(\mathcal{K}(G))$ for the continuous map induced by Ψ^H and $\check{\psi}^H = \mathrm{Spc}(\check{\Psi}^H): V_{G//H} \xrightarrow{v} \mathrm{Spc}(\mathcal{K}(G//H)) \xrightarrow{\psi^H} \mathrm{Spc}(\mathcal{K}(G))$ for its restriction to the cohomological open of $G//H$. If we need to specify the ambient group we write $\psi^{H;G}$ for ψ^H , etc. We saw in [BG22, Section 6] that ψ^H is a closed map, and a closed immersion if $H \trianglelefteq G$ is normal. Every prime $\mathcal{P} \in \mathrm{Spc}(\mathcal{K}(G))$ is of the form $\mathcal{P} = \mathcal{P}_G(H, \mathfrak{p}) := \check{\psi}^H(\mathfrak{p})$ for a p -subgroup $H \leq G$ and a point $\mathfrak{p} \in V_{G//H}$ in the cohomological open of the Weyl group of H , in a unique way up to G -conjugation; see [BG22, Theorem 6.16]. Hence the pieces $V_G(H) := \check{\psi}(V_{G//H})$ yield a partition $\mathrm{Spc}(\mathcal{K}(G)) = \sqcup_{H \in \mathrm{Sub}_p(G)/_G} V_G(H)$ into relatively open strata $V_G(H)$, homeomorphic to $V_{G//H}$. The crux of the problem is to understand how these strata $V_G(H) \simeq V_{G//H}$ attach together topologically, to build the space $\mathrm{Spc}(\mathcal{K}(G))$.

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2. THE COLIMIT THEOREM

To reduce the determination of $\mathrm{Spc}(\mathcal{K}(G))$ to the elementary abelian case, we invoke the *category $\mathcal{E}_p(G)$ of elementary abelian p -sections* of a finite group G . Recall that a section of G is a pair (H, K) of subgroups with K normal in H .

2.1. *Construction.* We denote by $\mathcal{E}_p(G)$ the category whose objects are pairs (H, K) where $K \trianglelefteq H$ are p -subgroups of G such that H/K is elementary abelian. Morphisms $(H, K) \rightarrow (H', K')$ are defined to be elements $g \in G$ such that

$$H^g \cap K' \leq K^g \leq H^g \leq H'.$$

Composition of morphisms is defined by multiplication in G .

2.2. *Examples.* Let us highlight three types of morphisms in $\mathcal{E}_p(G)$.

(a) We have an isomorphism $g: (H, K) \xrightarrow{\sim} (H^g, K^g)$ in $\mathcal{E}_p(G)$ for every $g \in G$. Intuitively, we can think of this as the group isomorphism $c_g: H/K \xrightarrow{\sim} H^g/K^g$.

(b) For every object (H', K') in $\mathcal{E}_p(G)$ and every subgroup $H \leq H'$, we have a well-defined object $(H, H \cap K')$ and the morphism $1: (H, H \cap K') \rightarrow (H', K')$. Intuitively, we think of it as the inclusion $H/(H \cap K') \hookrightarrow H'/K'$ of a subgroup.

(c) For (H, K) in $\mathcal{E}_p(G)$ and a subgroup $\bar{L} = L/K$ of H/K , for $K \leq L \leq H$, there is another morphism in $\mathcal{E}_p(G)$ associated to $1 \in G$, namely $1: (H, L) \rightarrow (H, K)$. This one does not correspond to an intuitive group homomorphism $H/L \twoheadrightarrow H/K$, as K is *smaller* than L . Instead, H/L is the quotient of H/K by $\bar{L} \trianglelefteq H/K$. This last morphism will be responsible for the modular $\bar{L}$ -fixed-points functor.

2.3. *Remark.* Every morphism $g: (H, K) \rightarrow (H', K')$ in $\mathcal{E}_p(G)$ is a composition of three morphisms of the above types (a), (b) and (c) in the following canonical way:

$$\begin{array}{ccccc} H & & H & & {}^g H' & & H' \\ \nabla & \xrightarrow{(c)} & \nabla & \xrightarrow{(b)} & \nabla & \xrightarrow{(a)} & \nabla \\ K & & H \cap {}^g K' & & {}^g K' & & K' \end{array}$$

where the first two are given by $1 \in G$ and the last is given by g . In particular, the rank of the elementary abelian group H/K increases or stays the same along any morphism $(H, K) \rightarrow (H', K')$ in this category, as this is true with (a), (b) and (c).

2.4. *Construction.* To every object (H, K) in $\mathcal{E}_p(G)$, we associate the tt-category $\mathcal{K}(H/K) = \mathbf{K}_b(\text{perm}(H/K; k))$. For every morphism $g: (H, K) \rightarrow (H', K')$ in $\mathcal{E}_p(G)$, we set $\bar{K} = K/(H \cap {}^g K')$ and we define a functor of tt-categories:

$$\mathcal{K}(g): \mathcal{K}(H'/K') \xrightarrow{c_g^*} \mathcal{K}({}^g H' / {}^g K') \xrightarrow{\text{Res}} \mathcal{K}(H/(H \cap {}^g K')) \xrightarrow{\Psi^{\bar{K}}} \mathcal{K}(H/K)$$

using that $H/(H \cap {}^g K')$ is a subgroup of ${}^g H' / {}^g K'$ for the restriction, and using that $(H/(H \cap {}^g K'))/\bar{K} = H/K$ for the modular fixed-points functor $\Psi^{\bar{K}}$.

It follows from [BG22, Proposition 4.15 and Corollary 4.18] that $\mathcal{K}(-)$ is a contravariant (pseudo) functor on $\mathcal{E}_p(G)$ with values in tt-categories:

$$(2.5) \quad \mathcal{K}: \mathcal{E}_p(G)^{\text{op}} \longrightarrow \text{tt-Cat}.$$

We can compose this with $\text{Spc}(-)$, which incidentally makes the coherence of the 2-isomorphisms accompanying (2.5) irrelevant, and obtain a covariant functor from $\mathcal{E}_p(G)$ to topological spaces. Let us compare this diagram of spaces (and its colimit) with the space $\text{Spc}(\mathcal{K}(G))$. For each $(H, K) \in \mathcal{E}_p(G)$, we have a tt-functor

$$(2.6) \quad \mathcal{K}(G) \xrightarrow{\text{Res}_H^G} \mathcal{K}(H) \xrightarrow{\Psi^K} \mathcal{K}(H/K)$$

which yields a natural transformation from the constant functor $(H, K) \mapsto \mathcal{K}(G)$ to the functor $\mathcal{K}: \mathcal{E}_p(G)^{\text{op}} \rightarrow \text{tt-Cat}$ of (2.5). The above Ψ^K is $\Psi^{K;H}$. Since $H \leq N_G K$, the tt-functor (2.6) is also $\text{Res}_{H/K}^{G/K} \circ \Psi^{K;G}: \mathcal{K}(G) \rightarrow \mathcal{K}(G//K) \rightarrow \mathcal{K}(H/K)$. Applying $\text{Spc}(-)$ to this observation, we obtain a commutative square:

$$(2.7) \quad \begin{array}{ccc} \text{Spc}(\mathcal{K}(H/K)) & \xrightarrow{\psi^{K;H}} & \text{Spc}(\mathcal{K}(H)) \\ \rho_{H/K} \downarrow & \searrow \varphi_{(H,K)} & \downarrow \rho_H \\ \text{Spc}(\mathcal{K}(G//K)) & \xrightarrow{\psi^{K;G}} & \text{Spc}(\mathcal{K}(G)) \end{array}$$

whose diagonal we baptize $\varphi_{(H,K)}$. In summary, we obtain a continuous map

$$(2.8) \quad \varphi: \text{colim}_{(H,K) \in \mathcal{E}_p(G)} \text{Spc}(\mathcal{K}(H/K)) \rightarrow \text{Spc}(\mathcal{K}(G))$$

whose component $\varphi_{(H,K)}$ at (H,K) is the diagonal map in (2.7).

2.9. Lemma. (a) *Each of the maps $\mathrm{Spc}(\mathcal{K}(g)) : \mathrm{Spc}(\mathcal{K}(H/K)) \rightarrow \mathrm{Spc}(\mathcal{K}(H'/K'))$ in the colimit diagram (2.8) is a closed immersion.*

(b) *Each of the components $\varphi_{(H,K)} : \mathrm{Spc}(\mathcal{K}(H/K)) \rightarrow \mathrm{Spc}(\mathcal{K}(G))$ of (2.8) is closed and preserves the dimension of points (i.e. the Krull dimension of their closure).*

Proof. These statements follow from two facts, see Recollection 1.8: When $N \trianglelefteq G$ is normal the map $\psi^N : \mathrm{Spc}(\mathcal{K}(G/N)) \hookrightarrow \mathrm{Spc}(\mathcal{K}(G))$ is a closed immersion. When $H \leq G$ is any subgroup, the map $\rho_H : \mathrm{Spc}(\mathcal{K}(H)) \rightarrow \mathrm{Spc}(\mathcal{K}(G))$ is closed, hence lifts specializations, and it moreover satisfies ‘Incomparability’ by [Bal16]. $\square$

We are now ready to prove Theorem 1.1:

2.10. Theorem. *For any finite group G , the map φ in (2.8) is a homeomorphism.*

Proof. Each component $\varphi_{(H,K)}$ is a closed map and thus φ is a closed map. For surjectivity, by Recollection 1.8, we know that $\mathrm{Spc}(\mathcal{K}(G))$ is covered by the subsets $\psi^K(V_{G//K})$, over all p -subgroups $K \leq G$. Hence it suffices to know that the $\mathrm{Im}(\rho_E)$ cover $V_{G//K} = \mathrm{Spc}(\mathrm{D}_b(G//K))$ as $E \leq G//K$ runs through all elementary p -subgroups. (Such an E must be of the form H/K for an object $(H,K) \in \mathcal{E}_p(G)$.) This holds by a classical result of Quillen [Qui71]; see [Bal16, Theorem 4.10].

The key point is injectivity. Take $\mathcal{P} \in \mathrm{Spc}(\mathcal{K}(H/K))$ and $\mathcal{P}' \in \mathrm{Spc}(\mathcal{K}(H'/K'))$ with same image in $\mathrm{Spc}(\mathcal{K}(G))$. Write $\mathcal{P} = \mathcal{P}_{H/K}(L/K, \mathfrak{p})$ for suitable arguments ($K \leq L \leq H$, $\mathfrak{p} \in V_{H/L}$) and note that the map induced by $1 : (H,L) \rightarrow (H,K)$ in $\mathcal{E}_p(G)$ sends $\mathcal{P}_{H/L}(1, \mathfrak{p}) \in \mathrm{Spc}(\mathcal{K}(H/L))$ to $\mathcal{P}$. So we may assume $L = K$. By [BG22, Remark 6.6], the image of $\mathcal{P} = \mathcal{P}_{H/K}(1, \mathfrak{p})$ in $\mathrm{Spc}(\mathcal{K}(G))$ is $\mathcal{P}_G(K, \bar{\rho}(\mathfrak{p}))$ where $\bar{\rho} : V_{H/K} \rightarrow V_{G//K}$ is induced by restriction. Similarly, we may assume $\mathcal{P}' = \mathcal{P}_{H'/K'}(1, \mathfrak{p}')$ for $\mathfrak{p}' \in V_{H'/K'}$ and we have $\mathcal{P}_G(K, \bar{\rho}(\mathfrak{p})) = \mathcal{P}_G(K', \bar{\rho}'(\mathfrak{p}'))$ in $\mathrm{Spc}(\mathcal{K}(G))$ and need to show that $\mathcal{P}$ and $\mathcal{P}'$ are identified in the colimit (2.8).

By [BG22, Theorem 6.16], the relation $\mathcal{P}_G(K, \bar{\rho}(\mathfrak{p})) = \mathcal{P}_G(K', \bar{\rho}'(\mathfrak{p}'))$ can only hold because of G -conjugation, meaning that there exists $g \in G$ such that $K' = K^g$ and $\bar{\rho}'(\mathfrak{p}') = \bar{\rho}(\mathfrak{p})^g$ in $V_{G//K'}$. Using the map $g : (H,K) \rightarrow (H^g, K^g)$ in $\mathcal{E}_p(G)$ we may replace $H, K, \mathfrak{p}$ by $H^g, K^g, \mathfrak{p}^g$ and reduce to the case $K = K'$. In other words, we have two points $\mathcal{P} = \mathcal{P}_{H/K}(1, \mathfrak{p}) \in \mathrm{Spc}(\mathcal{K}(H/K))$ and $\mathcal{P}' = \mathcal{P}_{H'/K}(1, \mathfrak{p}') \in \mathrm{Spc}(\mathcal{K}(H'/K))$ corresponding to two p -subgroups $H, H' \leq G$ containing the *same* normal subgroup K and two cohomological primes $\mathfrak{p} \in V_{H/K}$ and $\mathfrak{p}' \in V_{H'/K}$ such that $\bar{\rho}(\mathfrak{p}) = \bar{\rho}'(\mathfrak{p}')$ in $V_{G//K}$ under the maps $\bar{\rho}$ and $\bar{\rho}'$ induced by restriction along $H/K \leq G//K$ and $H'/K \leq G//K$ respectively.

If we let $\bar{G} = G//K = (N_G K)/K$, we have two elementary abelian p -subgroups $\bar{H} = H/K$ and $\bar{H}' = H'/K$ of $\bar{G}$, each with a point in their cohomological open, $\mathfrak{p} \in V_{\bar{H}}$ and $\mathfrak{p}' \in V_{\bar{H}'}$, and those two points have the same image in the cohomological open $V_{\bar{G}}$ of the ‘ambient’ group $\bar{G}$. By Quillen [Qui71] (or [Bal16, §4]) again, we know that this coalescence must happen because of an element $\bar{g} \in \bar{G}$, that is, a $g \in N_G K$, and a prime $\mathfrak{q} \in V_{\bar{H} \cap {}^g \bar{H}'}$ that maps to $\mathfrak{p}$ and to $\mathfrak{p}'$ under the maps $V_{\bar{H} \cap {}^g \bar{H}'} \rightarrow V_{\bar{H}}$ and $V_{\bar{H} \cap {}^g \bar{H}'} \rightarrow V_{\bar{H}'}$ respectively. But our category $\mathcal{E}_p(G)$ contains all such conjugation-inclusion morphisms coming from the orbit category of G . Specifically, we have two morphisms $1 : (H \cap {}^g H', K) \rightarrow (H, K)$ and $g : (H \cap {}^g H', K) \rightarrow (H', K)$ in $\mathcal{E}_p(G)$, under which the point $\mathcal{P}_{(H \cap {}^g H')/K}(1, \mathfrak{q})$ maps to $\mathcal{P}_{H/K}(1, \mathfrak{p}) = \mathcal{P}$ and $\mathcal{P}_{H'/K}(1, \mathfrak{p}') = \mathcal{P}'$ respectively. This shows that $\mathcal{P} = \mathcal{P}'$ in the domain of (2.8) as required. $\square$

2.11. *Remark.* By [BG22, Proposition 8.1], the space $\mathrm{Spc}(\mathcal{K}(G))$ is noetherian. Hence the topology is entirely characterized by the inclusion of primes. Now, suppose that $\mathcal{P}$ is the image under $\varphi_{(H,K)}: \mathrm{Spc}(\mathcal{K}(E)) \rightarrow \mathrm{Spc}(\mathcal{K}(G))$ of some $\mathcal{P}' \in \mathrm{Spc}(\mathcal{K}(E))$ for an elementary abelian subquotient $E = H/K$ corresponding to a section $(H, K) \in \mathcal{E}_p(G)$. Then the only way for another prime $\mathcal{Q} \in \mathrm{Spc}(\mathcal{K}(G))$ to belong to the closure of $\mathcal{P}$ is to be itself the image of some point $\mathcal{Q}'$ of $\mathrm{Spc}(\mathcal{K}(E))$ in the closure of $\mathcal{P}'$. This follows from Lemma 2.9. In other words, the question of inclusion of primes can also be reduced to the elementary abelian case.

3. INVERTIBLE OBJECTS AND TWISTED COHOMOLOGY

In this section we introduce a graded ring whose homogeneous spectrum helps us understand the topology on $\mathrm{Spc}(\mathcal{K}(G))$, at least for G elementary abelian. This graded ring, called the *twisted cohomology ring* (Definition 3.16), consists of morphisms between $\mathbb{1}$ and certain invertible objects. It all starts in the cyclic case.

3.1. *Example.* Let $C_p = \langle \sigma \mid \sigma^p = 1 \rangle$ be the cyclic group of prime order p , with a chosen generator. We write $kC_p = k[\sigma]/(\sigma^p - 1)$ as $k[\tau]/\tau^p$ for $\tau = \sigma - 1$. Then the coaugmentation and augmentation maps become:

$$\eta: k \xrightarrow{1 \mapsto \tau^{p-1}} kC_p \quad \text{and} \quad \epsilon: kC_p \xrightarrow{\tau \mapsto 0} k.$$

For p odd, we denote the first terms of the ‘standard’ minimal resolution of k by

$$u_p = (0 \rightarrow kC_p \xrightarrow{\tau} kC_p \xrightarrow{\epsilon} k \rightarrow 0).$$

We view this in $\mathcal{K}(C_p)$ with k in homological degree zero. One can verify directly that u_p is $\otimes$ -invertible, with $u_p^{\otimes -1} = u_p^\vee \cong (0 \rightarrow k \xrightarrow{\eta} kC_p \xrightarrow{\tau} kC_p \rightarrow 0)$. Alternatively, one can use the conservative pair of functors $\mathbb{F}^H: \mathcal{K}(C_p) \rightarrow \mathrm{D}_b(k)$ for $H \in \{C_p, 1\}$, corresponding to the only closed points $\mathcal{M}(C_p)$ and $\mathcal{M}(1)$ of $\mathrm{Spc}(\mathcal{K}(C_p))$. Those functors map u_p to the $\otimes$ -invertibles k and $k[2]$ in $\mathrm{D}_b(k)$, respectively.

For $p = 2$, we have a similar but shorter $\otimes$ -invertible object in $\mathcal{K}(C_2)$

$$u_2 = (0 \rightarrow kC_2 \xrightarrow{\epsilon} k \rightarrow 0)$$

again with k in degree zero.

3.2. *Notation.* To avoid constantly distinguishing cases, we abbreviate

$$2' := \begin{cases} 2 & \text{if } p > 2 \\ 1 & \text{if } p = 2. \end{cases}$$

For any finite group G and any index- p normal subgroup N , we can inflate the $\otimes$ -invertible u_p of Example 3.1 along $\pi: G \twoheadrightarrow G/N \simeq C_p$ to a $\otimes$ -invertible in $\mathcal{K}(G)$.

3.3. *Definition.* Let $N \triangleleft G$ be a normal subgroup of index p . We define

$$(3.4) \quad u_N := \begin{cases} \cdots \rightarrow 0 \rightarrow k(G/N) \xrightarrow{\tau} k(G/N) \xrightarrow{\epsilon} k \rightarrow 0 \rightarrow \cdots & \text{if } p \text{ is odd} \\ \cdots \rightarrow 0 \rightarrow 0 \rightarrow k(G/N) \xrightarrow{\epsilon} k \rightarrow 0 \rightarrow \cdots & \text{if } p = 2 \end{cases}$$

with k in degree zero. We also define two morphisms

$$a_N: \mathbb{1} \rightarrow u_N \quad \text{and} \quad b_N: \mathbb{1} \rightarrow u_N[-2']$$

as follows. The morphism a_N is the identity in degree zero, independently of p :

$$\begin{array}{ccccccc} \mathbb{1} & = & \cdots & \longrightarrow & 0 & \longrightarrow & k \rightarrow 0 \rightarrow \cdots \\ a_N \downarrow & & & & \downarrow & & \downarrow 1 \\ u_N & = & \cdots & \longrightarrow & k(G/N) & \xrightarrow{\epsilon} & k \rightarrow 0 \rightarrow \cdots \end{array}$$

The morphism b_N is given by $\eta: k \rightarrow k(G/N)$ in degree zero, as follows:

$$\begin{array}{ccc} \mathbb{1} & = & k \longrightarrow 0 \\ b_N \downarrow & & \downarrow \eta \quad \downarrow \\ u_N[-1] & = & k(G/N) \xrightarrow{\epsilon} k \end{array} \quad \text{and} \quad \begin{array}{ccc} \mathbb{1} & = & k \longrightarrow 0 \longrightarrow 0 \\ b_N \downarrow & & \downarrow \eta \quad \downarrow \\ u_N[-2] & = & k(G/N) \xrightarrow{\tau} k(G/N) \xrightarrow{\epsilon} k \end{array}$$

where the target u_N is shifted once to the right for $p = 2$ (as in the left-hand diagram above) and shifted twice for $p > 2$ (as in the right-hand diagram).

When p is odd there is furthermore a third morphism $c_N: \mathbb{1} \rightarrow u_N[-1]$, that is defined to be $\eta: k \rightarrow k(G/N)$ in degree zero. This c_N will play a lesser role.

In statements made for all primes p , simply ignore c_N in the case $p = 2$ (or think $c_N = 0$). Here is an example of such a statement, whose meaning should now be clear: *The morphisms a_N and b_N , and c_N (for p odd), are inflated from G/N .*

3.5. Remark. Technically, u_N depends not only on an index- p subgroup $N \triangleleft G$ but also on the choice of a generator of G/N , to identify G/N with C_p . If one needs to make this distinction, one can write u_π for a chosen epimorphism $\pi: G \twoheadrightarrow C_p$. This does not change the isomorphism type of u_N , namely $\ker(\pi) = \ker(\pi')$ implies $u_\pi \cong u_{\pi'}$. (We expand on this topic in Remark 8.2.)

3.6. Lemma. *Let $N \triangleleft G$ be a normal subgroup of index p and let $q \geq 1$. Then there is a canonical isomorphism in $\mathcal{K}(G)$*

$$u_N^{\otimes q} \cong (\cdots 0 \rightarrow k(G/N) \xrightarrow{\tau} k(G/N) \xrightarrow{\tau^{p-1}} \cdots \xrightarrow{\tau} k(G/N) \xrightarrow{\epsilon} k \rightarrow 0 \cdots)$$

where the first $k(G/N)$ sits in homological degree $2' \cdot q$ and k sits in degree 0.

Proof. It is an exercise over the cyclic group C_p . Then inflate along $G \twoheadrightarrow G/N$. $\square$

3.7. Remark. The morphism $b_N: \mathbb{1} \rightarrow u_N[-2']$ of Definition 3.3 is a quasi-isomorphism and the fraction

$$\zeta_N := (b_N[2'])^{-1} \circ a_N: \mathbb{1} \rightarrow u_N \leftarrow \mathbb{1}[2']$$

is a well-known morphism $\zeta_N \in \text{Hom}_{\text{D}_b(kG)}(\mathbb{1}, \mathbb{1}[2']) = H^{2'}(G, k)$ in the derived category $\text{D}_b(kG)$. For G elementary abelian, these ζ_N generate the cohomology k -algebra $H^*(G, k)$, on the nose for $p = 2$ and modulo nilpotents for p odd.

We sometimes write $\zeta_N^+ = \frac{a_N}{b_N}$ for ζ_N in order to distinguish it from the inverse fraction $\zeta_N^- := \frac{b_N}{a_N}$ that exists wherever a_N is inverted. Of course, when both a_N and b_N are inverted, we have $\zeta_N^- = (\zeta_N^+)^{-1} = \zeta_N^{-1}$.

3.8. Remark. The switch of factors (12): $u_N \otimes u_N \cong u_N \otimes u_N$ can be computed directly to be the identity (over C_p , then inflate). Alternatively, it must be multiplication by a square-one element of $\text{Aut}(\mathbb{1}) = k^\times$, hence ± 1 . One can then apply the tensor-functor $\Psi^G: \mathcal{K}(G) \rightarrow \text{D}_b(k)$, under which u_N goes to $\mathbb{1}$, to rule out -1 .

It follows that for p odd, $u_N[-1]$ has switch -1 , and consequently every morphism $\mathbb{1} \rightarrow u_N[-1]$ must square to zero. In particular $c_N \otimes c_N = 0$. This nilpotence explains why c_N will play no significant role in the topology.

We can describe the image under modular fixed-points functors of the $\otimes$ -invertible objects u_N and of the morphisms a_N and b_N . (We leave c_N as an exercise.)

3.9. Proposition. *Let $H \trianglelefteq G$ be a normal p -subgroup. Then for every index- p normal subgroup $N \triangleleft G$, we have in $\mathcal{K}(G/H)$*

$$\Psi^H(u_N) \cong \begin{cases} u_{N/H} & \text{if } H \leq N \\ \mathbb{1} & \text{if } H \not\leq N \end{cases}$$

and under this identification

$$\Psi^H(a_N) = \begin{cases} a_{N/H} & \text{if } H \leq N \\ 1_{\mathbb{1}} & \text{if } H \not\leq N \end{cases} \quad \text{and} \quad \Psi^H(b_N) = \begin{cases} b_{N/H} & \text{if } H \leq N \\ 0 & \text{if } H \not\leq N. \end{cases}$$

Proof. Direct from Definition 3.3 and $\Psi^H(k(X)) \cong k(X^H)$ for $X = G/N$. $\square$

For restriction, there is an analogous pattern but with the cases ‘swapped’.

3.10. Proposition. *Let $H \leq G$ be a subgroup. Then for every index- p normal subgroup $N \triangleleft G$, we have in $\mathcal{K}(H)$*

$$\text{Res}_H^G(u_N) \cong \begin{cases} \mathbb{1}[2'] & \text{if } H \leq N \\ u_{N \cap H} & \text{if } H \not\leq N \end{cases}$$

and under this identification

$$\text{Res}_H^G(a_N) = \begin{cases} 0 & \text{if } H \leq N \\ a_{N \cap H} & \text{if } H \not\leq N \end{cases} \quad \text{and} \quad \text{Res}_H^G(b_N) = \begin{cases} 1_{\mathbb{1}} & \text{if } H \leq N \\ b_{N \cap H} & \text{if } H \not\leq N. \end{cases}$$

Proof. Direct from Definition 3.3 and the Mackey formula for $\text{Res}_H^G(k(G/N))$. $\square$

We can combine the above two propositions and handle Ψ^H for non-normal H , since by definition $\Psi^{H;G} = \Psi^{H;N_G H} \circ \text{Res}_{N_G H}^G$. Here is an application of this.

3.11. Corollary. *Let $H \leq G$ be a p -subgroup and $N \triangleleft G$ of index p . Recall the ‘residue’ tt -functor $\mathbb{F}^H = \text{Res}_1 \circ \Psi^H : \mathcal{K}(G) \rightarrow \text{D}_b(k)$ at the closed point $\mathcal{M}(H)$.*

- (a) *If $H \not\leq N$ then $\mathbb{F}^H(a_N)$ is an isomorphism.*
- (b) *If $H \leq N$ then $\mathbb{F}^H(b_N)$ is an isomorphism.*

Proof. We apply Proposition 3.10 for $N_G H \leq G$ and Proposition 3.9 for $H \triangleleft N_G H$. For (a), $H \not\leq N$ forces $N_G H \not\leq N$ and $H \not\leq N \cap N_G H$. Hence $\Psi^H(a_N) = \Psi^{H;N_G H} \text{Res}_{N_G H}^G(a_N) = \Psi^{H;N_G H}(a_{N \cap N_G H}) = 1_{\mathbb{1}}$ is an isomorphism. Similarly for (b), if $N_G H \leq N$ then $\Psi^H(b_N)$ is an isomorphism and if $N_G H \not\leq N$ it is the quasi-isomorphism $b_{(N \cap N_G H)/H}$. Thus $\mathbb{F}^H(b_N)$ is an isomorphism in $\text{D}_b(k)$. $\square$

Let us now prove that the morphisms a_N and b_N , and c_N (for p odd), generate all morphisms from the unit $\mathbb{1}$ to tensor products of u_N ’s. This is a critical fact.

3.12. Lemma. *Let $N_1, \dots, N_\ell$ be index- p normal subgroups of G and abbreviate $u_i := u_{N_i}$ for $i = 1, \dots, \ell$ and similarly $a_i := a_{N_i}$ and $b_i := b_{N_i}$ and $c_i := c_{N_i}$ (see Definition 3.3). Let $q_1, \dots, q_\ell \in \mathbb{N}$ be non-negative integers and $s \in \mathbb{Z}$. Then every morphism $f : \mathbb{1} \rightarrow u_1^{\otimes q_1} \otimes \dots \otimes u_\ell^{\otimes q_\ell}[s]$ in $\mathcal{K}(G)$ is a k -linear combination of tensor products of (i.e. a ‘polynomial’ in) the morphisms a_i and b_i , and c_i (for p odd).*

Proof. We proceed by induction on ℓ . The case $\ell = 0$ is just $\text{End}_{\mathcal{K}(G)}^\bullet(\mathbb{1}) = k$. Suppose $\ell \geq 1$ and the result known for $\ell - 1$. Up to reducing to $\ell - 1$, we can assume that the $N_1, \dots, N_\ell$ are all distinct. Set for readability

$$v := u_1^{\otimes q_1} \otimes \dots \otimes u_{\ell-1}^{\otimes q_{\ell-1}}[s], \quad N := N_\ell, \quad u := u_\ell = u_N \quad \text{and} \quad q := q_\ell$$

so that f is a morphism of the form

$$f: \mathbb{1} \rightarrow v \otimes u^{\otimes q}.$$

We then proceed by induction on $q \geq 0$. We assume the result known for $q - 1$ (the case $q = 0$ holds by induction hypothesis on ℓ). The proof will now depend on p .

Suppose first that $p = 2$. Consider the exact triangle in $\mathcal{K}(G)$

$$(3.13) \quad \begin{array}{ccccccc} u^{\otimes(q-1)} & = & \dots 0 & \longrightarrow & 0 & \longrightarrow & k(G/N) \xrightarrow{\tau} \dots \xrightarrow{\tau} k(G/N) \xrightarrow{\epsilon} k \rightarrow 0 \dots \\ a_N \downarrow & & & & \downarrow & & \parallel \\ u^{\otimes q} & = & \dots 0 & \rightarrow & k(G/N) \xrightarrow{\tau} k(G/N) \xrightarrow{\tau} \dots \xrightarrow{\tau} k(G/N) \xrightarrow{\epsilon} k \rightarrow 0 \dots \\ \downarrow & & & & \parallel & & \downarrow \\ k(G/N)[q] & = & \dots 0 & \rightarrow & k(G/N) \longrightarrow 0 \longrightarrow \dots \longrightarrow 0 \longrightarrow 0 \rightarrow 0 \dots \end{array}$$

where k is in degree zero. (See Lemma 3.6.) Tensoring the above triangle with v and applying $\text{Hom}_G(\mathbb{1}, -) := \text{Hom}_{\mathcal{K}(G)}(\mathbb{1}, -)$ we get an exact sequence

$$(3.14) \quad \begin{array}{ccc} \text{Hom}_G(\mathbb{1}, v \otimes u^{\otimes(q-1)}) & \xrightarrow{a_N} & \text{Hom}_G(\mathbb{1}, v \otimes u^{\otimes q}) \longrightarrow \text{Hom}_G(\mathbb{1}, v \otimes k(G/N)[q]) \\ & \downarrow \scriptstyle f & \parallel \\ & & f' \in \text{Hom}_N(\mathbb{1}, \text{Res}_N^G(v)[q]) \end{array}$$

Our morphism f belongs to the middle group. By adjunction, the right-hand term is $\text{Hom}_N(\mathbb{1}, \text{Res}_N^G(v)[q])$. Now since all $N_1, \dots, N_\ell = N$ are distinct, we can apply Proposition 3.10 to compute $\text{Res}_N^G(v)$ and we know by induction hypothesis (on ℓ) that the image f' of our f in this group $\text{Hom}_N(\mathbb{1}, \text{Res}_N^G(v)[q])$ is a k -linear combination of tensor products of $a_{N_i \cap N}$ and $b_{N_j \cap N}$ for $1 \leq i, j \leq \ell - 1$, performed over the group N . We can perform the ‘same’ k -linear combination of tensor products of a_i ’s and b_j ’s over the group G , thus defining a morphism $f'' \in \text{Hom}_G(\mathbb{1}, v[q])$. We can now multiply f'' with $b_N^{\otimes q}: \mathbb{1} \rightarrow u_N^{\otimes q}[-q]$ to obtain a morphism $f''b_N^q$ in the same group $\text{Hom}_G(\mathbb{1}, v \otimes u^{\otimes q})$ that contains f . Direct computation shows that the image of this $f''b_N^q$ in $\text{Hom}_N(\mathbb{1}, \text{Res}_N(v)[s])$ is also equal to f' . The key point is that $b_N^{\otimes q}$ is simply $\eta: k \rightarrow k(G/N)$ in degree q and this η is also the unit of the $\text{Res}_N^G \dashv \text{Ind}_N^G$ adjunction. In other words, the difference $f - f''b_N^q$ comes from the left-hand group $\text{Hom}_G(\mathbb{1}, v \otimes u^{\otimes(q-1)})$ in the exact sequence (3.14), reading

$$f = f''b_N^q + f'''a_N$$

for some $f''' \in \text{Hom}_G(\mathbb{1}, v \otimes u^{\otimes(q-1)})$. By induction hypothesis (on q), f''' is a polynomial in a_i ’s and b_j ’s. Since f'' also was such a polynomial, so is f .

The proof for p odd follows a similar pattern of induction on q , with one complication. The cone of the canonical map $a_N: u_N^{\otimes(q-1)} \rightarrow u_N^{\otimes q}$ is not simply $k(G/N)$ in a single degree as in (3.13) but rather the complex

$$C := (\dots \rightarrow 0 \rightarrow k(G/N) \xrightarrow{\tau} k(G/N) \rightarrow 0 \rightarrow \dots)$$

with $k(G/N)$ in two consecutive degrees $2q$ and $2q - 1$. So the exact sequence

$$(3.15) \quad \mathrm{Hom}_G(\mathbb{1}, v \otimes u^{\otimes(q-1)}) \xrightarrow{a_N} \mathrm{Hom}_G(\mathbb{1}, v \otimes u^{\otimes q}) \longrightarrow \mathrm{Hom}_G(\mathbb{1}, v \otimes C)$$

has a more complicated third term than the one of (3.14). That third term $\mathrm{Hom}_G(\mathbb{1}, v \otimes C)$ itself fits in its own exact sequence associated to the exact triangle $k(G/N)[2q - 1] \xrightarrow{\tau} k(G/N)[2q - 1] \rightarrow C \rightarrow k(G/N)[2q]$. Each of the terms $\mathrm{Hom}_G(\mathbb{1}, v \otimes k(G/N)[*]) \cong \mathrm{Hom}_N(\mathbb{1}, \mathrm{Res}_N^G(v)[*])$ can be computed as before, by adjunction. The image of f in $\mathrm{Hom}_N(\mathbb{1}, \mathrm{Res}_N^G(v)[2q])$ can again be lifted to a polynomial $f' b_N^q : \mathbb{1} \rightarrow v \otimes u^{\otimes q}$ so that the image of the difference $f - f' b_N^q$ in $\mathrm{Hom}_G(\mathbb{1}, v \otimes C)$ comes from some element in $\mathrm{Hom}_N(\mathbb{1}, \mathrm{Res}_N^G(v)[2q - 1])$. That element may be lifted to a polynomial $f'' b_N^{q-1} c_N : \mathbb{1} \rightarrow v \otimes u^{\otimes q}$, and we obtain

$$f = f' b_N^q + f'' b_N^{q-1} c_N + f''' a_N$$

for some $f''' \in \mathrm{Hom}_G(\mathbb{1}, v \otimes u^{\otimes(q-1)})$ similarly as before. $\square$

We can now assemble all the hom groups of Lemma 3.12 into a big graded ring.

3.16. *Definition.* We denote the set of all index- p normal subgroups of G by

$$(3.17) \quad \mathcal{N} = \mathcal{N}(G) := \{ N \triangleleft G \mid [G:N] = p \}.$$

Let $\mathbb{N}^{\mathcal{N}} = \mathbb{N}^{\mathcal{N}(G)} = \{ q : \mathcal{N}(G) \rightarrow \mathbb{N} \}$ be the monoid of *twists*, i.e. tuples of non-negative integers indexed by this finite set. Consider the $(\mathbb{Z} \times \mathbb{N}^{\mathcal{N}})$ -graded ring

$$(3.18) \quad \mathbf{H}^{\bullet\bullet}(G) = \mathbf{H}^{\bullet\bullet}(G; k) := \bigoplus_{s \in \mathbb{Z}} \bigoplus_{q \in \mathbb{N}^{\mathcal{N}}} \mathrm{Hom}_{\mathcal{K}(G)} \left(\mathbb{1}, \bigotimes_{N \in \mathcal{N}} u_N^{\otimes q(N)} [s] \right).$$

Its multiplication is induced by the tensor product in $\mathcal{K}(G)$. We call $\mathbf{H}^{\bullet\bullet}(G)$ the *(permutation) twisted cohomology ring of G* . It is convenient to simply write

$$(3.19) \quad \mathbb{1}(q) = \bigotimes_{N \in \mathcal{N}} (u_N)^{\otimes q(N)}$$

for every twist $q \in \mathbb{N}^{\mathcal{N}(G)}$ and thus abbreviate $\mathbf{H}^{s,q}(G) = \mathrm{Hom}(\mathbb{1}, \mathbb{1}(q)[s])$.

3.20. *Remark.* The graded ring $\mathbf{H}^{\bullet\bullet}(G)$ is graded-commutative by using only the parity of the shift, not the twist; see Remark 3.8. In other words, we have

$$h_1 \cdot h_2 = (-1)^{s_1 \cdot s_2} h_2 \cdot h_1 \quad \text{when} \quad h_i \in \mathbf{H}^{s_i, q_i}(G).$$

For instance, for p odd, when dealing with the morphisms a_N and b_N , which land in even shifts of the object u_N , we do not have to worry too much about the order. This explains the ‘unordered’ notation $\zeta_N = \frac{a_N}{b_N}$ used in Remark 3.7.

The critical Lemma 3.12 gives the main property of this construction:

3.21. **Theorem.** *The twisted cohomology ring $\mathbf{H}^{\bullet\bullet}(G)$ of Definition 3.16 is a k -algebra generated by the finitely many elements a_N and b_N , and c_N (for p odd), of Definition 3.3, over all $N \triangleleft G$ of index p . In particular $\mathbf{H}^{\bullet\bullet}(G)$ is noetherian. $\square$*

3.22. *Example.* The reader can verify by hand that $\mathbf{H}^{\bullet\bullet}(C_2) = k[a_N, b_N]$, without relations, and that $\mathbf{H}^{\bullet\bullet}(C_p) = k[a_N, b_N, c_N] / \langle c_N^2 \rangle$ for p odd, where in both cases $N = 1$ is the only $N \in \mathcal{N}(C_p)$. This example is deceptive, for the $\{a_N, b_N, c_N\}_{N \in \mathcal{N}}$ usually satisfy some relations, as the reader can already check for $G = C_2 \times C_2$ for instance. We systematically discuss these relations in Section 8.

We conclude this section with some commentary.

3.23. *Remark.* The name ‘cohomology’ in Definition 3.16 is used in the loose sense of a graded endomorphism ring of the unit in a tensor-triangulated category. However, since we are using the tt-category $\mathcal{K}(G)$ and not $D_b(kG)$, the ring $H^{\bullet\bullet}(G)$ is quite different from $H^\bullet(G, k)$ in general. In fact, $H^{\bullet\bullet}(G)$ could even be rather dull. For instance, if G is a non-cyclic simple group then $\mathcal{N}(G) = \emptyset$ and $H^{\bullet\bullet}(G) = k$. We will make serious use of $H^{\bullet\bullet}(G)$ in Section 6 to describe $\mathrm{Spc}(\mathcal{K}(G))$ for G elementary abelian. In that case, $H^\bullet(G, k)$ is a localization of $H^{\bullet\bullet}(G)$. See Example 5.13.

3.24. *Remark.* By Proposition 3.9, there is no ‘collision’ in the twists: If there is an isomorphism $\mathbb{1}(q)[s] \simeq \mathbb{1}(q')[s']$ in $\mathcal{K}(G)$ then we must have $q = q'$ in $\mathbb{N}^\mathbb{N}$ and $s = s'$ in $\mathbb{Z}$. The latter is clear from $\mathbb{F}^G(u_N) \cong \mathbb{1}$ in $D_b(k)$, independently of N . We then conclude from $\mathbb{F}^N(\mathbb{1}(q)) \simeq \mathbb{1}[2'q(N)]$ in $D_b(k)$, for each $N \in \mathcal{N}$.

3.25. *Remark.* We only use positive twists $q(N)$ in (3.18). The reader can verify that already for $G = C_p$ cyclic, the $\mathbb{Z}^2$ -graded ring $\bigoplus_{(s,q) \in \mathbb{Z}^2} \mathrm{Hom}(\mathbb{1}, u_p^{\otimes q}[s])$ is not noetherian. See for instance [DHM24] for $p = 2$. However, negatively twisted elements tend to be nilpotent. So the $\mathbb{Z} \times \mathbb{Z}^\mathbb{N}$ -graded version of $H^{\bullet\bullet}(G)$ may yield the same topological information as our $\mathbb{Z} \times \mathbb{N}^\mathbb{N}$ -graded one. We have not pushed this investigation of negative twists, as it brought no benefit to our analysis.

4. AN OPEN COVER OF THE SPECTRUM

In this section, we extract some topological information about $\mathrm{Spc}(\mathcal{K}(G))$ from the twisted cohomology ring $H^{\bullet\bullet}(G)$ of Definition 3.16 and the maps a_N and b_N of Definition 3.3, associated to every index- p normal subgroup N in $\mathcal{N} = \mathcal{N}(G)$.

Recall from [BG22, Construction 2.14] that we can use tensor-induction to associate to every subgroup $H \leq G$ a Koszul object $\mathrm{kos}_G(H) = \bigotimes \mathrm{Ind}_H^G(0 \rightarrow k \xrightarrow{1} k \rightarrow 0)$. It generates in $\mathcal{K}(G)$ the tt-ideal $\mathrm{Ker}(\mathrm{Res}_H^G)$, see [BG22, Proposition 2.21]:

$$(4.1) \quad \langle \mathrm{kos}_G(H) \rangle_{\mathcal{K}(G)} = \mathrm{Ker}(\mathrm{Res}_H^G: \mathcal{K}(G) \rightarrow \mathcal{K}(H)).$$

4.2. **Lemma.** *Let $N \triangleleft G$ be a normal subgroup of index p . Then we have:*

- (a) *In $\mathcal{K}(G)$, the object $\mathrm{cone}(a_N)$ generates the same thick subcategory as $k(G/N)$. In particular, $\mathrm{supp}(\mathrm{cone}(a_N)) = \mathrm{supp}(k(G/N))$.*
- (b) *In $\mathcal{K}(G)$, the object $\mathrm{cone}(b_N)$ generates the same thick tensor-ideal as $\mathrm{kos}_G(N)$. In particular, $\mathrm{supp}(\mathrm{cone}(b_N)) = \mathrm{supp}(\mathrm{kos}_G(N)) = \mathrm{supp}(\mathrm{Ker}(\mathrm{Res}_N^G))$.*

Proof. For $p = 2$, we have $\mathrm{cone}(a_N) = k(G/N)[1]$ so the first case is clear. For p odd, we have $\mathrm{cone}(a_N)[-1] \simeq (0 \rightarrow k(G/N) \xrightarrow{\tau} k(G/N) \rightarrow 0) = \mathrm{cone}(\tau|_{k(G/N)})$. Hence $\mathrm{cone}(a_N) \in \mathrm{thick}(k(G/N))$. Conversely, since $\tau^p = 0$, the octahedron axiom inductively shows that $k(G/N) \in \mathrm{thick}(\mathrm{cone}(\tau|_{k(G/N)}))$. This settles (a).

For (b), the complex $s := \mathrm{cone}(b_N)[2']$ becomes split exact when restricted to N since it is inflated from an exact complex on G/N . In degree one we have $s_1 = k(G/N)$, whereas $s_0 = k$. Hence [BG22, Corollary 2.20] tells us that the complex s generates the tt-ideal $\mathrm{Ker}(\mathrm{Res}_N^G: \mathcal{K}(G) \rightarrow \mathcal{K}(N))$. We conclude by (4.1). $\square$

4.3. **Corollary.** *Let $N \triangleleft G$ be of index p . Then $\mathrm{cone}(a_N) \otimes \mathrm{cone}(b_N) = 0$.*

Proof. By Lemma 4.2 it suffices to show $k(G/N) \otimes \mathrm{kos}_G(N) = 0$. By Frobenius, this follows from $\mathrm{Res}_N^G(\mathrm{kos}_G(N)) = 0$, which holds by (4.1). $\square$

We now relate the spectrum of $\mathcal{K}(G)$ to the homogeneous spectrum of $H^{\bullet\bullet}(G)$, in the spirit of [Bal10]. The comparison map of [Bal10] is denoted by $\rho^\bullet$ but we prefer a more descriptive notation (and here, the letter ρ is reserved for $\mathrm{Spc}(\mathrm{Res})$).

4.4. Proposition. *There is a continuous ‘comparison’ map*

$$\mathrm{comp}_G: \mathrm{Spc}(\mathcal{K}(G)) \longrightarrow \mathrm{Spec}^h(H^{\bullet\bullet}(G))$$

mapping a tt-prime $\mathcal{P}$ to the ideal generated by those homogeneous $f \in H^{\bullet\bullet}(G)$ whose cone does not belong to $\mathcal{P}$. It is characterized by the fact that for all f

$$(4.5) \quad \mathrm{comp}_G^{-1}(Z(f)) = \mathrm{supp}(\mathrm{cone}(f)) = \{ \mathcal{P} \mid f \text{ is not invertible in } \mathcal{K}(G)/\mathcal{P} \}$$

where $Z(f) = \{ \mathfrak{p} \mid f \in \mathfrak{p} \}$ is the closed subset of $\mathrm{Spec}^h(H^{\bullet\bullet}(G))$ defined by f .

Proof. The fact that the homogeneous ideal $\mathrm{comp}_G(\mathcal{P})$ is prime comes from [Bal10, Theorem 4.5]. Equation (4.5) is essentially a reformulation of the definition. $\square$

4.6. Remark. The usual notation for $Z(f)$ would be $V(f)$, and $D(f)$ for its open complement. Here, we already use V for V_G and for $V_G(H)$, and the letter D is certainly overworked in our trade. So we stick to $Z(f)$ and $Z(f)^c$.

4.7. Notation. In view of Proposition 4.4, for any f , the open subset of $\mathrm{Spc}(\mathcal{K}(G))$

$$(4.8) \quad \mathrm{open}(f) := \mathrm{open}(\mathrm{cone}(f)) = \{ \mathcal{P} \mid f \text{ is invertible in } \mathcal{K}(G)/\mathcal{P} \}$$

is the preimage by $\mathrm{comp}_G: \mathrm{Spc}(\mathcal{K}(G)) \rightarrow \mathrm{Spec}^h(H^{\bullet\bullet}(G))$ of the principal open $Z(f)^c = \{ \mathfrak{p} \mid f \notin \mathfrak{p} \}$. It is the open locus of $\mathrm{Spc}(\mathcal{K}(G))$ where f is invertible. In particular, our distinguished elements a_N and b_N (see Definition 3.3) give us the following open subsets of $\mathrm{Spc}(\mathcal{K}(G))$, for every $N \in \mathcal{N}(G)$:

$$\mathrm{open}(a_N) = \mathrm{comp}_G^{-1}(Z(a_N)^c), \text{ the open where } a_N \text{ is invertible, and}$$

$$\mathrm{open}(b_N) = \mathrm{comp}_G^{-1}(Z(b_N)^c), \text{ the open where } b_N \text{ is invertible.}$$

Since $(c_N)^2 = 0$ by Remark 3.8, we do not have much use for $\mathrm{open}(c_N) = \emptyset$.

4.9. Corollary. *With notation as above, we have for every $N \triangleleft G$ of index p*

$$\mathrm{open}(a_N) \cup \mathrm{open}(b_N) = \mathrm{Spc}(\mathcal{K}(G)).$$

Proof. We compute $\mathrm{open}(a_N) \cup \mathrm{open}(b_N) = \mathrm{open}(\mathrm{cone}(a_N)) \cup \mathrm{open}(\mathrm{cone}(b_N)) = \mathrm{open}(\mathrm{cone}(a_N) \otimes \mathrm{cone}(b_N)) = \mathrm{open}(0_{\mathcal{K}(E)}) = \mathrm{Spc}(\mathcal{K}(G))$, using Corollary 4.3. $\square$

4.10. Remark. Every object u_N is not only $\otimes$ -invertible in $\mathcal{K}(G)$ but actually locally trivial over $\mathrm{Spc}(\mathcal{K}(G))$, which is a stronger property in general tt-geometry. Indeed, Corollary 4.9 tells us that around each point of $\mathrm{Spc}(\mathcal{K}(G))$, either u_N becomes isomorphic to $\mathbb{1}$ via a_N , or u_N becomes isomorphic to $\mathbb{1}[2']$ via b_N . This holds for *one* invertible u_N . We now construct a fine enough open cover of $\mathrm{Spc}(\mathcal{K}(G))$ such that *every* u_N is trivialized on each open.

4.11. Proposition. *Let $H \leq G$ be a p -subgroup. Define an open of $\mathrm{Spc}(\mathcal{K}(G))$ by*

$$(4.12) \quad U(H) = U_G(H) := \bigcap_{\substack{N \in \mathcal{N} \\ H \not\leq N}} \mathrm{open}(a_N) \cap \bigcap_{\substack{N \in \mathcal{N} \\ H \leq N}} \mathrm{open}(b_N).$$

Then the closed point $\mathcal{M}(H) \in \mathrm{Spc}(\mathcal{K}(G))$ belongs to this open $U(H)$. Consequently $\{U(H)\}_{H \in \mathrm{Sub}_p(G)}$ is an open cover of $\mathrm{Spc}(\mathcal{K}(G))$.

Proof. The point $\mathcal{M}(H) = \text{Ker}(\mathbb{F}^H)$ belongs to $U(H)$ by Corollary 3.11. It follows by general tt-geometry that $\{U(H)\}_H$ is a cover: Let $\mathcal{P} \in \text{Spc}(\mathcal{K}(G))$; there exists a closed point in $\overline{\{\mathcal{P}\}}$, that is, some $\mathcal{M}(H)$ that admits $\mathcal{P}$ as a generalization; but then $\mathcal{M}(H) \in U(H)$ forces $\mathcal{P} \in U(H)$ since open subsets are generalization-closed. $\square$

For a p -group, we now discuss $U(H)$ at the two extremes $H = 1$ and $H = G$.

4.13. Recollection. Let G be a p -group and $F = F(G) = \cap_{N \in \mathcal{N}(G)} N$ be its Frattini subgroup. So $F \triangleleft G$ and G/F is the largest elementary abelian quotient of G .

4.14. Proposition. *Let G be a p -group with Frattini subgroup F . The closed complement of the open $U_G(1)$ is the support of $\text{kos}_G(F)$, i.e. the closed support of the tt-ideal $\text{Ker}(\text{Res}_F^G)$ of $\mathcal{K}(G)$. In particular, if G is elementary abelian then $U_G(1)$ is equal to the cohomological open $V_G = \text{Spc}(\text{D}_b(kG)) \cong \text{Spec}^h(\mathbf{H}^\bullet(G, k))$.*

Proof. By definition, $U(1) = \cap_{N \in \mathcal{N}} \text{open}(b_N)$. By Lemma 4.2, its closed complement is $\cup_{N \in \mathcal{N}} \text{supp}(\text{kos}_G(N))$. By [BG22, Corollary 6.17], for every $K \leq G$

$$(4.15) \quad \text{supp}(\text{kos}_G(K)) = \{ \mathcal{P}(H, \mathfrak{p}) \mid H \not\leq_G K \}$$

(taking all possible $\mathfrak{p} \in V_{G//H}$). It follows that our closed complement of $U(1)$ is

$$\begin{aligned} \cup_{N \in \mathcal{N}(G)} \text{supp}(\text{kos}_G(N)) &\stackrel{(4.15)}{=} \{ \mathcal{P}(H, \mathfrak{p}) \mid \exists N \in \mathcal{N}(G) \text{ such that } H \not\leq_G N \} \\ &= \{ \mathcal{P}(H, \mathfrak{p}) \mid H \not\leq \cap_{N \in \mathcal{N}(G)} N \} = \{ \mathcal{P}(H, \mathfrak{p}) \mid H \not\leq F \} \stackrel{(4.15)}{=} \text{supp}(\text{kos}_G(F)). \end{aligned}$$

The statement with $\text{Ker}(\text{Res}_F^G)$ then follows from (4.1). Finally, if G is elementary abelian then $F = 1$ and $\text{Ker}(\text{Res}_1^G) = \mathcal{K}_{\text{ac}}(G)$ is the tt-ideal of acyclic complexes. The complement of its support is $\text{Spc}(\mathcal{K}(G)/\mathcal{K}_{\text{ac}}(G)) = \text{Spc}(\text{D}_b(kG)) = V_G$. $\square$

In the above proof, we showed that $\cup_{N \in \mathcal{N}} \text{supp}(\text{kos}(N)) = \text{supp}(\text{kos}(F))$ thanks to the fact that $\cap_{N \in \mathcal{N}} N = F$. So the very same argument gives us:

4.16. Corollary. *Let G be a p -group and let $N_1, \dots, N_r \in \mathcal{N}(G)$ be some index- p subgroups such that $N_1 \cap \dots \cap N_r$ is the Frattini subgroup F . (This can be realized with r equal to the p -rank of G/F .) Then $U_G(1) = \cap_{i=1}^r \text{open}(b_{N_i})$ already. Hence if $\mathcal{P} \in \text{open}(b_{N_i})$ for all $i = 1, \dots, r$ then $\mathcal{P} \in \text{open}(b_N)$ for all $N \in \mathcal{N}(G)$. $\square$*

Let us turn to the open $U_G(H)$ for the p -subgroup at the other end: $H = G$.

4.17. Proposition. *Let G be a p -group. Then the complement of the open $U_G(G)$ is the union of the images of the spectra $\text{Spc}(\mathcal{K}(H))$ under the maps $\rho_H = \text{Spc}(\text{Res}_H)$, over all the proper subgroups $H \leq G$.*

Proof. By Lemma 4.2, the closed complement of $U(G) = \cap_{N \in \mathcal{N}} \text{open}(a_N)$ equals $\cup_{N \in \mathcal{N}} \text{supp}(k(G/N))$. For every $H \leq G$, we have $\text{supp}(k(G/H)) = \text{Im}(\rho_H)$; see [BG22, Proposition 3.7] if necessary. This gives the result because restriction to any proper subgroup factors via some index- p subgroup, since G is a p -group. $\square$

4.18. Remark. Let G be a p -group. This open complement $U(G)$ of $\cup_{H \leq G} \text{Im}(\rho_H)$ could be called the ‘geometric open’. Indeed, the localization functor

$$\Phi^G: \mathcal{K}(G) \rightarrow \frac{\mathcal{K}(G)}{\langle k(G/H) \mid H \leq G \rangle}$$

corresponding to $U(G)$ is analogous to the way the geometric fixed-points functor is constructed in topology. For more on this topic, see [BG22, Remark 3.11].

4.19. *Remark.* For G not a p -group, the open $U(G)$ is not defined (we assume $H \in \text{Sub}_p(G)$ in Proposition 4.11) and the ‘geometric open’ is void anyway as we have $\text{Im}(\rho_P) = \text{Spc}(\mathcal{K}(G))$ for any p -Sylow $P \not\leq G$. The strategy to analyze non- p -groups is to first descend to the p -Sylow, using that Res_P is faithful.

4.20. *Remark.* We saw in Proposition 4.17 that the complement of $U(G)$ is covered by the images of the closed maps $\rho_H = \text{Spc}(\text{Res}_H)$ for $H \not\leq G$. We could wonder whether another closed map into $\text{Spc}(\mathcal{K}(G))$ covers $U(G)$ itself. The answer is the closed immersion $\psi^F: \text{Spc}(\mathcal{K}(G/F)) \hookrightarrow \text{Spc}(\mathcal{K}(G))$ induced by the modular fixed-points functor Ψ^F with respect to the Frattini subgroup $F \triangleleft G$. This can be deduced from the results of Section 2 or verified directly, as we now outline. Indeed, every prime $\mathcal{P} = \mathcal{P}_G(K, \mathfrak{p})$ for $K \leq G$ and $\mathfrak{p} \in V_{G//K}$ comes by Quillen from some elementary abelian subgroup $E = H/K \leq G//K = (N_G K)/K$. One verifies that unless $N_G K = G$ and $H = G$, the prime $\mathcal{P}$ belongs to the image of $\rho_{G'}$ for a proper subgroup G' of G . Thus if $\mathcal{P}$ belongs to $U(G)$, we must have $E = H/K = G/K$ for $K \triangleleft G$. Such a K must contain the Frattini and the result follows.

5. TWISTED COHOMOLOGY UNDER TT-FUNCTORS

Still for a general finite group G , we gather some properties of the twisted cohomology ring $H^{\bullet\bullet}(G)$ introduced in Definition 3.16. We describe its behavior under specific tt-functors, namely restriction, modular fixed-points and localization onto the open subsets $U_G(H)$. Recall that $\mathcal{N} = \mathcal{N}(G) = \{N \triangleleft G \mid [G:N] = p\}$.

5.1. *Remark.* Twisted cohomology $H^{\bullet\bullet}(G)$ is graded over a monoid of the form $\mathbb{Z} \times \mathbb{N}^\ell$. The ring homomorphisms induced by the above tt-functors will be homogeneous *with respect to a certain homomorphism* γ on the corresponding grading monoids, meaning of course that the image of a homogeneous element of degree (s, q) is homogeneous of degree $\gamma(s, q)$. The ‘shift’ part (in $\mathbb{Z}$) is rather straightforward. The ‘twist’ part (in $\mathbb{N}^\ell$) will depend on the effect of said tt-functors on the u_N .

Let us start with modular fixed-points, as they are relatively easy.

5.2. *Construction.* Let $H \triangleleft G$ be a normal subgroup. By Proposition 3.9, the tt-functor $\Psi^H: \mathcal{K}(G) \rightarrow \mathcal{K}(G/H)$ maps every u_N for $N \not\geq H$ to $\mathbb{1}$, whereas it maps u_N for $N \geq H$ to $u_{N/H}$. This defines a homomorphism of grading monoids

$$(5.3) \quad \gamma = \gamma_{\Psi^H}: \mathbb{Z} \times \mathbb{N}^{\mathcal{N}(G)} \rightarrow \mathbb{Z} \times \mathbb{N}^{\mathcal{N}(G/H)}$$

given by $\gamma(s, q) = (s, \bar{q})$ where $\bar{q}(N/H) = q(N)$ for every $N/H \in \mathcal{N}(G/H)$. In other words, $q \mapsto \bar{q}$ is simply restriction $\mathbb{N}^{\mathcal{N}(G)} \rightarrow \mathbb{N}^{\mathcal{N}(G/H)}$ along the canonical inclusion $\mathcal{N}(G/H) \hookrightarrow \mathcal{N}(G)$. By Proposition 3.9, for every twist $q \in \mathbb{N}^{\mathcal{N}(G)}$, we have a canonical isomorphism $\Psi^H(\mathbb{1}(q)) \cong \mathbb{1}(\bar{q})$. Therefore the modular fixed-points functor Ψ^H defines a ring homomorphism also denoted

$$(5.4) \quad \begin{array}{ccc} \Psi^H: & H^{\bullet\bullet}(G) & \longrightarrow H^{\bullet\bullet}(G/H) \\ & \left(\mathbb{1} \xrightarrow{f} \mathbb{1}(q)[s] \right) & \longmapsto \left(\mathbb{1} \xrightarrow{\Psi^H(f)} \Psi^H(\mathbb{1}(q)[s]) \cong \mathbb{1}(\bar{q})[s] \right) \end{array}$$

which is homogeneous with respect to γ_{Ψ^H} in (5.3).

Restriction is a little more subtle, as some twists pull-back to non-trivial shifts.

5.5. *Construction.* Let $\alpha: G' \rightarrow G$ be a group homomorphism. Restriction along α defines a tt-functor $\alpha^* = \text{Infl}_{G'}^{\text{Im } \alpha} \circ \text{Res}_{\text{Im } \alpha}^G: \mathcal{K}(G) \rightarrow \mathcal{K}(\text{Im } \alpha) \rightarrow \mathcal{K}(G')$. Combining Proposition 3.10 for $\text{Res}_{\text{Im } \alpha}^G$ with the obvious behavior of the u_N under inflation (by construction), we see that $\alpha^*(u_N) \cong \mathbb{1}[2']$ if $N \geq \text{Im } \alpha$ and $\alpha^*(u_N) \cong u_{\alpha^{-1}(N)}$ if $N \not\geq \text{Im } \alpha$ (which is equivalent to $\alpha^{-1}(N) \in \mathcal{N}(G')$). Hence for every $(s, q) \in \mathbb{Z} \times \mathbb{N}^{\mathcal{N}(G)}$ we have a canonical isomorphism $\alpha^*(\mathbb{1}(q)[s]) \cong \mathbb{1}(q')[s']$ where $s' = s + 2' \sum_{N \geq \text{Im } \alpha} q(N)$ and $q': \mathcal{N}(G') \rightarrow \mathbb{N}$ is defined for every $N' \in \mathcal{N}(G')$ as

$$q'(N') = \sum_{N \in \mathcal{N}(G) \text{ s.t. } \alpha^{-1}(N) = N'} q(N).$$

(In particular $q'(N') = 0$ if $N' \not\geq \ker(\alpha)$.) These formulas define a homomorphism $(s, q) \mapsto (s', q')$ of abelian monoids that we denote

$$(5.6) \quad \gamma = \gamma_{\alpha^*}: \mathbb{Z} \times \mathbb{N}^{\mathcal{N}(G)} \rightarrow \mathbb{Z} \times \mathbb{N}^{\mathcal{N}(G')}.$$

The restriction functor α^* defines a ring homomorphism

$$(5.7) \quad \begin{aligned} \alpha^*: \quad & \mathbf{H}^{\bullet\bullet}(G) \longrightarrow \mathbf{H}^{\bullet\bullet}(G') \\ & (\mathbb{1} \xrightarrow{f} \mathbb{1}(q)[s]) \longmapsto (\mathbb{1} \xrightarrow{\alpha^*(f)} \alpha^*(\mathbb{1}(q)[s]) \cong \mathbb{1}(q')[s']) \end{aligned}$$

which is homogeneous with respect to γ_{α^*} in (5.6).

5.8. *Remark.* For instance, $\alpha: G \rightarrow G/H$ can be the quotient by a normal subgroup $H \trianglelefteq G$. In that case α^* is inflation, which is a section of modular fixed-points Ψ^H . It follows that the homomorphism Ψ^H in (5.4) is split surjective. (This also means that the composed effect on gradings $\gamma_{\Psi^H} \circ \gamma_{\alpha^*} = \text{id}$ is trivial.)

Without changing the group G , we can also localize the twisted cohomology ring $\mathbf{H}^{\bullet\bullet}(G)$ by restricting to an open $U(H)$ of $\text{Spc}(\mathcal{K}(G))$, as defined in Proposition 4.11. Recall the elements $a_N, b_N \in \mathbf{H}^{\bullet\bullet}(G)$ from Definition 3.3.

5.9. *Definition.* Let $H \leq G$ be a p -subgroup. Let $S_H \subset \mathbf{H}^{\bullet\bullet}(G)$ be the multiplicative subset of the graded ring $\mathbf{H}^{\bullet\bullet}(G)$ generated by all a_N such that $H \not\leq N$ and all b_N such that $H \leq N$, for all $N \in \mathcal{N}(G)$. Recall that the a_N and b_N are central by Remark 3.20. We define a $\mathbb{Z}$ -graded ring

$$(5.10) \quad \mathcal{O}_G^\bullet(H) := (\mathbf{H}^{\bullet\bullet}(G)[S_H^{-1}])_{0\text{-twist}}$$

as the twist-zero part of the localization of $\mathbf{H}^{\bullet\bullet}(G)$ with respect to S_H . Explicitly, the homogeneous elements of $\mathcal{O}_G^\bullet(H)$ consist of fractions $\frac{f}{g}$ where $f, g \in \mathbf{H}^{\bullet\bullet}(G)$ are such that $g: \mathbb{1} \rightarrow \mathbb{1}(q)[t]$ is a product of the chosen a_N, b_N in S_H , meaning that $\mathbb{1}(q)[t]$ is the $\otimes$ -product of the corresponding u_N for a_N and $u_N[-2']$ for b_N , whereas $f: \mathbb{1} \rightarrow \mathbb{1}(q)[s]$ is any morphism in $\mathcal{K}(G)$ with the same $\mathcal{N}$ -twist q as the denominator. Thus $\mathcal{O}_G^\bullet(H)$ is $\mathbb{Z}$ -graded by the shift only: The degree of $\frac{f}{g}$ is the difference $s - t$ between the shifts of f and g .

5.11. *Remark.* It follows from Lemma 3.12 (and Remark 3.24) that the $\mathbb{Z}$ -graded ring $\mathcal{O}_G^\bullet(H)$ is generated as a k -algebra by the elements

$$\{ \zeta_N^+, \xi_N^+ \mid H \leq N \} \cup \{ \zeta_N^-, \xi_N^- \mid H \not\leq N \}$$

where $\zeta_N^+ = a_N/b_N$ is of degree $+2'$ and $\zeta_N^- = b_N/a_N$ of degree $-2'$ as in Remark 3.7, and where (only for p odd) the additional elements $\xi_N^\pm$ are $\xi_N^+ := c_N/b_N$ of degree $+1$, and $\xi_N^- := c_N/a_N$ of degree -1 . (For $p = 2$, simply ignore the $\xi_N^\pm$.) In

general, all these elements satisfy some relations; see Theorem 8.13. Beware that here ξ_N^- is never the inverse of ξ_N^+ . In fact, both are nilpotent.

In fact, we can perform the central localization of the whole category $\mathcal{K}(G)$

$$\mathcal{L}(H) = \mathcal{L}_G(H) := \mathcal{K}(G)[S_H^{-1}]$$

with respect to the central multiplicative subset S_H of Definition 5.9.

5.12. *Construction.* The tt-category $\mathcal{L}(H) = \mathcal{K}(G)[S_H^{-1}]$ has the same objects as $\mathcal{K}(G)$ and morphisms $x \rightarrow y$ of the form $\frac{f}{g}$ where $g: \mathbb{1} \rightarrow u$ belongs to S_H , for u a tensor-product of shifts of u_N 's according to g (as in Definition 5.9) and where $f: x \rightarrow u \otimes y$ is any morphism in $\mathcal{K}(G)$ with ‘same’ twist u as the denominator g . This category $\mathcal{K}(G)[S_H^{-1}]$ is also the Verdier quotient of $\mathcal{K}(G)$ by the tt-ideal $\langle \{ \text{cone}(g) \mid g \in S_H \} \rangle$ and the above fraction $\frac{f}{g}$ corresponds to the Verdier fraction $x \xrightarrow{f} u \otimes y \xleftarrow{g \otimes \mathbb{1}} y$. See [Bal10, § 3] if necessary.

The $\mathbb{Z}$ -graded endomorphism ring $\text{End}_{\mathcal{L}(H)}^\bullet(\mathbb{1})$ of the unit in $\mathcal{L}(H) = \mathcal{K}(G)[S_H^{-1}]$ is thus the $\mathbb{Z}$ -graded ring $(S_H^{-1} H^{\bullet\bullet}(G))_{0\text{-twist}} = \mathcal{O}_G^\bullet(H)$ of Definition 5.9.

There is a general localization $\mathcal{K}|_U$ of a tt-category $\mathcal{K}$ over a quasi-compact open $U \subseteq \text{Spc}(\mathcal{K})$ with closed complement Z . It is defined as $\mathcal{K}|_U = (\mathcal{K}/\mathcal{K}_Z)^\natural$. If we apply this to $U = U(H)$, we deduce from (4.12) that $U = \bigcap_{g \in S_H} \text{open}(g)$ has closed complement $Z = \bigcup_{g \in S_H} \text{supp}(\text{cone}(g))$ whose tt-ideal $\mathcal{K}(G)_Z$ is the above $\langle \{ \text{cone}(g) \mid g \in S_H \} \rangle$. In other words, the idempotent-completion of our $\mathcal{L}_G(H) = \mathcal{K}(G)[S_H^{-1}]$ is exactly $\mathcal{K}(G)|_{U(H)}$. As with any localization, we know that $\text{Spc}(\mathcal{L}_G(H))$ is a subspace of $\text{Spc}(\mathcal{K}(G))$, given here by $U = \bigcap_{g \in S_H} \text{open}(g) = U(H)$.

5.13. *Example.* For $G = E$ elementary abelian and the subgroup $H = 1$, the category $\mathcal{L}_E(1) = \mathcal{K}(E)|_{U(1)}$ in Construction 5.12 is simply the derived category $\mathcal{L}_E(1) = \text{D}_b(E)$, by Proposition 4.14. In that case, $\mathcal{O}_E^\bullet(1) \cong H^\bullet(E; k)$ is the actual cohomology ring of E . Since $H = 1 \leq N$ for all N , we are inverting all the b_N and no a_N . As noted in Corollary 4.16, we obtain the same ring (the cohomology of E) as soon as we invert enough $b_{N_1}, \dots, b_{N_r}$, namely, as soon as $N_1 \cap \dots \cap N_r = 1$.

We again obtain an induced homomorphism of multi-graded rings.

5.14. *Construction.* Let $H \leq G$ be a p -subgroup and consider the above central localization $(-)|_{U(H)}: \mathcal{K}(G) \rightarrow \mathcal{L}_G(H)$. As explained in Remark 4.10, the morphisms a_N and b_N give us explicit isomorphisms $(u_N)|_{U(H)} \cong \mathbb{1}$ if $N \not\geq H$ and $(u_N)|_{U(H)} \cong \mathbb{1}[2']$ if $N \geq H$. This yields a homomorphism on the grading

$$(5.15) \quad \gamma = \gamma_{U(H)}: \mathbb{Z} \times \mathbb{N}^{N(G)} \rightarrow \mathbb{Z}$$

defined by $\gamma(s, q) = s + 2' \sum_{N \geq H} q(N)$ and we obtain a ring homomorphism

$$(5.16) \quad (-)|_{U(H)}: H^{\bullet\bullet}(G) \longrightarrow \text{End}_{\mathcal{L}_G(H)}^\bullet(\mathbb{1}) = \mathcal{O}_G^\bullet(H)$$

which is homogeneous with respect to the homomorphism $\gamma_{U(H)}$ of (5.15).

5.17. *Remark.* It is easy to verify that the continuous maps induced on homogeneous spectra by the ring homomorphisms constructed above are compatible with the comparison map of Proposition 4.4. In other words, if $F: \mathcal{K}(G) \rightarrow \mathcal{K}(G')$ is a tt-functor and if the induced homomorphism $F: H^{\bullet\bullet}(G) \rightarrow H^{\bullet\bullet}(G')$ is homogenous

with respect to $\gamma = \gamma_F: \mathbb{Z} \times \mathbb{N}^{(G)} \rightarrow \mathbb{Z} \times \mathbb{N}^{(G')}$, for instance $F = \Psi^H$ or $F = \alpha^*$ as in Constructions 5.2 and 5.5, then the following square commutes:

$$(5.18) \quad \begin{array}{ccc} \mathrm{Spc}(\mathcal{K}(G')) & \xrightarrow{\mathrm{Spc}(F)} & \mathrm{Spc}(\mathcal{K}(G)) \\ \downarrow \mathrm{comp}_{G'} & & \downarrow \mathrm{comp}_G \\ \mathrm{Spec}^h(\mathbf{H}^{\bullet\bullet}(G')) & \xrightarrow{\mathrm{Spec}^h(F)} & \mathrm{Spec}^h(\mathbf{H}^{\bullet\bullet}(G)). \end{array}$$

This follows from $F(\mathrm{cone}(f)) \simeq \mathrm{cone}(F(f))$ in $\mathcal{K}(G')$ for any $f \in \mathbf{H}^{\bullet\bullet}(G')$.

5.19. *Remark.* Similarly, for every $H \in \mathrm{Sub}_p(G)$ the following square commutes

$$(5.20) \quad \begin{array}{ccc} U_G(H) = \mathrm{Spc}(\mathcal{L}_G(H)) & \hookrightarrow & \mathrm{Spc}(\mathcal{K}(G)) \\ \downarrow \mathrm{comp}_{\mathcal{L}(H)} & & \downarrow \mathrm{comp}_G \\ \mathrm{Spec}^h(\mathcal{O}_G^\bullet(H)) & \hookrightarrow & \mathrm{Spec}^h(\mathbf{H}^{\bullet\bullet}(G)) \end{array}$$

where the left-hand vertical map is the classical comparison map of [Bal10] for the tt-category $\mathcal{L}_G(H)$ and the $\otimes$ -invertible $\mathbb{1}[1]$. The horizontal inclusions are the ones corresponding to the localizations with respect to S_H , as in Constructions 5.12 and 5.14. In fact, it is easy to verify that the square (5.20) is cartesian, in view of $U_G(H) = \bigcap_{g \in S_H} \mathrm{open}(g) = \bigcap_{g \in S_H} \mathrm{comp}_G^{-1}(Z(g)^c)$ by Construction 5.12 and (4.5).

We can combine the above functors. Here is a useful example.

5.21. **Proposition.** *Let $H \trianglelefteq G$ be a normal subgroup such that G/H is elementary abelian. Then we have a commutative square*

$$(5.22) \quad \begin{array}{ccc} V_{G/H} = \mathrm{Spc}(\mathrm{D}_b(k(G/H))) & \xrightarrow{\check{\Psi}^H} & \mathrm{Spc}(\mathcal{K}(G)) \\ \downarrow \mathrm{comp}_{\mathrm{D}_b(k(G/H))} \simeq & & \downarrow \mathrm{comp}_G \\ \mathrm{Spec}^h(\mathbf{H}^\bullet(G/H, k)) & \hookrightarrow & \mathrm{Spec}^h(\mathbf{H}^{\bullet\bullet}(G)) \end{array}$$

and in particular, its diagonal $\mathrm{comp}_G \circ \check{\Psi}^H$ is injective.

Proof. The functor $\check{\Psi}^H: \mathcal{K}(G) \rightarrow \mathrm{D}_b(k(G/H))$ is the modular fixed-points functor $\Psi^H: \mathcal{K}(G) \rightarrow \mathcal{K}(G/H)$ composed with $\Upsilon_{G/H}: \mathcal{K}(G/H) \rightarrow \mathrm{D}_b(k(G/H))$, which is the central localization $(-)|_{U(1)}$ over the cohomological open, by Proposition 4.14; see Example 5.13. Thus we obtain two commutative squares (5.20) and (5.18):

$$\begin{array}{ccccc} \mathrm{Spc}(\mathrm{D}_b(k(G/H))) & \xrightarrow{v_{G/H}} & \mathrm{Spc}(\mathcal{K}(G/H)) & \xrightarrow{\psi^H} & \mathrm{Spc}(\mathcal{K}(G)) \\ \downarrow \mathrm{comp}_{\mathrm{D}_b(k(G/H))} \simeq & & \downarrow \mathrm{comp}_{G/H} & & \downarrow \mathrm{comp}_G \\ \mathrm{Spec}^h(\mathbf{H}^\bullet(G/H, k)) & \hookrightarrow & \mathrm{Spec}^h(\mathbf{H}^{\bullet\bullet}(G/H)) & \hookrightarrow & \mathrm{Spec}^h(\mathbf{H}^{\bullet\bullet}(G)) \end{array}$$

the left-hand one for the central localization of $\mathcal{K}(G/H)$ over the open $U_{G/H}(1) = V_{G/H}$, and the right-hand one for the tt-functor $\Psi^H: \mathcal{K}(G) \rightarrow \mathcal{K}(G/H)$. Note that the bottom-right map is injective because the ring homomorphism in question, $\Psi^H: \mathbf{H}^{\bullet\bullet}(G) \rightarrow \mathbf{H}^{\bullet\bullet}(G/H)$ defined in (5.4), is surjective by Remark 5.8. $\square$

6. THE ELEMENTARY ABELIAN CASE

In this central section, we apply the general constructions of Sections 3 to 5 in the case of $G = E$ elementary abelian. We start with a key fact that is obviously wrong in general (e.g. for a non-cyclic simple group, the target space is just a point).

6.1. Proposition. *Let E be an elementary abelian group. The comparison map*

$$\text{comp}_E: \text{Spc}(\mathcal{K}(E)) \rightarrow \text{Spec}^h(\mathbf{H}^{\bullet\bullet}(E))$$

of Proposition 4.4 is injective.

Proof. Let $H, N \leq E$ with $[E: N] = p$. Suppose first that $H \not\leq N$. We use the map $\check{\psi}^H = \text{Spc}(\check{\Psi}^H): V_{E/H} \rightarrow \text{Spc}(\mathcal{K}(E))$ of Recollection 1.8. Then

$$\begin{aligned} (\check{\psi}^H)^{-1}(\text{open}(b_N)) &= (\check{\psi}^H)^{-1}(\text{open}(\text{cone}(b_N))) && \text{by definition, see (4.8)} \\ &= \text{open}(\text{cone}(\check{\Psi}^H(b_N))) && \text{by general tt-geometry} \\ &= \text{open}(\text{cone}(0: \mathbb{1} \rightarrow \mathbb{1})) && \text{by Proposition 3.9} \\ &= \text{open}(\mathbb{1} \oplus \mathbb{1}[1]) = \emptyset. \end{aligned}$$

Thus $\text{Im}(\check{\psi}^H)$ does not meet $\text{open}(b_N)$ when $H \not\leq N$. Suppose now that $H \leq N$. A similar computation as above shows that $(\check{\psi}^H)^{-1}(\text{open}(b_N)) = \text{Spc}(\text{D}_b(k(E/H)))$ since in that case $\check{\Psi}^H(b_N)$ is an isomorphism in $\text{D}_b(k(E/H))$. Therefore $\text{Im}(\check{\psi}^H) \subseteq \text{open}(b_N)$ when $H \leq N$. Combining both observations, we have

$$(6.2) \quad \text{Im}(\check{\psi}^H) \cap \text{open}(b_N) \neq \emptyset \iff H \leq N.$$

Let now $\mathcal{P}, \mathcal{Q} \in \text{Spc}(\mathcal{K}(E))$ be such that $\text{comp}_E(\mathcal{P}) = \text{comp}_E(\mathcal{Q})$ in $\text{Spec}^h(\mathbf{H}^{\bullet\bullet}(E))$. Say $\mathcal{P} = \mathcal{P}_E(H, \mathfrak{p})$ and $\mathcal{Q} = \mathcal{P}_E(K, \mathfrak{q})$ for $H, K \leq E$ and $\mathfrak{p} \in V_{E/H}$ and $\mathfrak{q} \in V_{E/K}$. (See Recollection 1.8.) The assumption $\text{comp}_E(\mathcal{P}) = \text{comp}_E(\mathcal{Q})$ implies that $\mathcal{P} \in \text{open}(f)$ if and only if $\mathcal{Q} \in \text{open}(f)$, for every $f \in \mathbf{H}^{\bullet\bullet}(E)$. In particular applying this to $f = b_N$, we see that for every index- p subgroup $N \triangleleft E$ we have $\mathcal{P} \in \text{open}(b_N)$ if and only if $\mathcal{Q} \in \text{open}(b_N)$. By (6.2), this means that for every $N \in \mathcal{N}(E)$ we have

$$H \leq N \iff K \leq N.$$

Since E is elementary abelian, this forces $H = K$. So we have two points $\mathfrak{p}, \mathfrak{q} \in V_{E/H}$ that go to the same image under $V_{E/H} \xrightarrow{\check{\psi}^H} \text{Spc}(\mathcal{K}(E)) \xrightarrow{\text{comp}_E} \text{Spec}^h(\mathbf{H}^{\bullet\bullet}(E))$ but we know that this map is injective by Proposition 5.21 for $G = E$. $\square$

In fact, we see that the open $U(H)$ of $\text{Spc}(\mathcal{K}(E))$ defined in Proposition 4.11 matches perfectly the open $\text{Spec}^h(\mathcal{O}_E^\bullet(H))$ of $\text{Spec}^h(\mathbf{H}^{\bullet\bullet}(E))$ in Definition 5.9.

6.3. Theorem. *Let E be an elementary abelian p -group. Let $H \leq E$ be a subgroup. Then the comparison map of Proposition 4.4 restricts to a homeomorphism*

$$\text{comp}_E: U(H) \xrightarrow{\sim} \text{Spec}^h(\mathcal{O}_E^\bullet(H))$$

where $\mathcal{O}_E^\bullet(H)$ is the $\mathbb{Z}$ -graded endomorphism ring of the unit $\mathbb{1}$ in the localization $\mathcal{L}_E(H)$ of $\mathcal{K}(E)$ over the open $U(H)$.

Proof. Recall the tt-category $\mathcal{L}(H) = \mathcal{L}_E(H) := \mathcal{K}(E)[S_H^{-1}]$ of Construction 5.12, where $S_H \subset \mathbf{H}^{\bullet\bullet}(E)$ is the multiplicative subset generated by the homogeneous elements $\{a_N \mid H \not\leq N\} \cup \{b_N \mid H \leq N\}$ of Definition 5.9. In view of Remark 5.19, it suffices to show that the map $\text{comp}_{\mathcal{L}(H)}: \text{Spc}(\mathcal{L}(H)) \rightarrow \text{Spec}^h(\mathcal{O}_E^\bullet(H))$ is a homeomorphism. We have injectivity by Proposition 6.1. We also know that $\mathcal{O}_E^\bullet(H)$ is

noetherian by Theorem 3.21. It follows from [Bal10] that $\text{comp}_{\mathcal{L}(H)}$ is surjective. Hence it is a continuous bijection and we only need to prove that it is a closed map.

We claim that $\mathcal{L}(H)$ is generated by its $\otimes$ -unit $\mathbb{1}$. Namely, let $\mathcal{J} = \text{thick}_{\mathcal{L}(H)}(\mathbb{1})$ be the thick subcategory of $\mathcal{L}(H)$ generated by $\mathbb{1}$ and let us see that $\mathcal{J} = \mathcal{L}(H)$. Observe that $\mathcal{J}$ is a sub-tt-category of $\mathcal{L}(H)$. Let $N \in \mathcal{N}$ be an index- p subgroup. We claim that $k(E/N)$ belongs to $\mathcal{J}$. If $N \not\geq H$, then a_N is inverted in $\mathcal{L}(H)$, so $k(E/N) = 0$ in $\mathcal{L}(H)$ by Lemma 4.2(a). If $N \geq H$, then $b_N: \mathbb{1} \rightarrow u_N[-2']$ is inverted, so $u_N \in \mathcal{J}$ and we conclude again by Lemma 4.2(a) since $a_N: \mathbb{1} \rightarrow u_N$ is now a morphism in $\mathcal{J}$. For a general proper subgroup $K < E$, the module $k(E/K)$ is a tensor product of $k(E/N)$ for some $N \in \mathcal{N}$. (Here we use E elementary abelian again.) Hence $k(E/K)$ also belongs to $\mathcal{J}$ as the latter is a sub-tt-category of $\mathcal{L}(H)$. In short $\mathcal{J}$ contains all generators $k(E/H)$ for $H \leq E$. Therefore $\mathcal{L}(H) = \mathcal{J}$ is indeed generated by its unit. It follows from this and from noetherianity of $\text{End}_{\mathcal{L}(H)}^\bullet(\mathbb{1}) = \mathcal{O}_E^\bullet(H)$ that $\text{Hom}_{\mathcal{L}(H)}^\bullet(x, y)$ is a finitely generated $\mathcal{O}_E^\bullet(H)$ -module for every $x, y \in \mathcal{L}(H)$. We conclude from a general tt-geometric fact, observed by Lau [Lau23, Proposition 2.4], that the map comp must then be closed. $\square$

6.4. Corollary. *Let E be an elementary abelian p -group. Let $\mathcal{O}_E^\bullet$ be the sheaf of $\mathbb{Z}$ -graded rings on $\text{Spc}(\mathcal{K}(E))$ obtained by sheafifying $U \mapsto \text{End}_{\mathcal{K}(E)|_U}^\bullet(\mathbb{1})$. Then $(\text{Spc}(\mathcal{K}(E)), \mathcal{O}_E^\bullet)$ is a Dirac scheme in the sense of [HP23].*

Proof. We identified an affine cover $\{U(H)\}_{H \leq E}$ in Theorem 6.3. $\square$

6.5. Remark. This result further justifies the notation for the ring $\mathcal{O}_E^\bullet(H)$ in Definition 5.9. Indeed, this $\mathcal{O}_E^\bullet(H)$ is also the ring of sections $\mathcal{O}_E^\bullet(U(H))$ of the $\mathbb{Z}$ -graded structure sheaf $\mathcal{O}_E^\bullet$ over the open $U(H)$ of Proposition 4.11.

6.6. Corollary. *Let E be an elementary abelian p -group. Then the comparison map of Proposition 4.4 is an open immersion. More precisely, it defines a homeomorphism between $\text{Spc}(\mathcal{K}(E))$ and the following open subspace of $\text{Spec}^h(\mathbf{H}^{\bullet\bullet}(E))$:*

$$(6.7) \quad \{ \mathfrak{p} \in \text{Spec}^h(\mathbf{H}^{\bullet\bullet}(E)) \mid \text{for all } N \triangleleft E \text{ of index } p \text{ either } a_N \notin \mathfrak{p} \text{ or } b_N \notin \mathfrak{p} \}.$$

Proof. By Proposition 6.1, the (continuous) comparison map is injective. Therefore, it being an open immersion can be checked locally on the domain. By Proposition 4.11, the open $U(H)$ form an open cover of $\text{Spc}(\mathcal{K}(E))$. Theorem 6.3 tells us that each $U(H)$ is homeomorphic to the following open of $\text{Spec}^h(\mathbf{H}^{\bullet\bullet}(E))$

$$U'(H) := \bigcap_{N \not\geq H} Z(a_N)^c \cap \bigcap_{N \geq H} Z(b_N)^c$$

(recall that $Z(f)^c = \{ \mathfrak{p} \mid f \notin \mathfrak{p} \}$ is our notation for a principal open). So it suffices to verify that the union $\cup_{H \leq E} U'(H)$, is the open subspace of the statement (6.7). Let $\mathfrak{p} \in U'(H)$ for some $H \leq E$ and let $N \in \mathcal{N}(E)$; then clearly either $N \not\geq H$ in which case $a_N \notin \mathfrak{p}$, or $N \geq H$ in which case $b_N \notin \mathfrak{p}$. Conversely let $\mathfrak{p}$ belong to the open (6.7) and define $H = \cap_{M \in \mathcal{N} \text{ s.t. } b_M \notin \mathfrak{p}} M$. We claim that $\mathfrak{p} \in U'(H)$. Let $N \in \mathcal{N}$. If $N \not\geq H$ then $b_N \in \mathfrak{p}$ by construction of H and therefore $a_N \notin \mathfrak{p}$. So the last thing we need to prove is that $N \geq H$ implies $b_N \notin \mathfrak{p}$. One should be slightly careful here, as H was defined as the intersection of the $M \in \mathcal{N}$ such that $b_M \notin \mathfrak{p}$, and certainly such M 's will contain H , but we need to see why *every* $N \geq H$ satisfies $b_N \notin \mathfrak{p}$. This last fact follows from Corollary 4.16 applied to E/H . $\square$

6.8. *Example.* Consider the spectrum of $\mathcal{K}(C_p)$ for the cyclic group C_p of order p . By Example 3.22, the reduced ring $\mathcal{O}_{C_p}^\bullet(1)_{\text{red}}$ is $k[\zeta^+]$ with $\zeta^+ = a/b$ in degree $2'$ while $\mathcal{O}_{C_p}^\bullet(C_p)_{\text{red}} = k[\zeta^-]$ with $\zeta^- = b/a$. (The former is also Example 5.13.) Each of these has homogeneous spectrum the Sierpiński space and we easily deduce that

$$(6.9) \quad \text{Spc}(\mathcal{K}(C_p)) = \begin{array}{c} \bullet \\ \textcolor{red}{\diagdown} \quad \textcolor{green}{\diagup} \\ U(C_p) \quad U(1) \end{array}$$

confirming the computation of $\text{Spc}(\mathcal{K}(C_{p^n}))$ in [BG22, Proposition 7.3] for $n = 1$.

We can also view this as an instance of Corollary 6.6. Namely, still by Example 3.22, the reduced ring $H^{\bullet\bullet}(C_p)_{\text{red}}$ is $k[a, b]$ with a in degree 0 and b in degree $-2'$. Its homogeneous spectrum has one more point at the top:

$$\text{Spec}^h(H^{\bullet\bullet}(C_p)) = \begin{array}{c} \bullet \\ \diagdown \quad \diagup \\ \textcolor{red}{\bullet} \quad \textcolor{green}{\bullet} \\ \textcolor{red}{Z(a)^c} \quad \textcolor{green}{Z(b)^c} \end{array}$$

and this superfluous closed point $\langle a, b \rangle$ lies outside of the open subspace (6.7).

6.10. *Remark.* Let $K \leq H \leq E$. The functor $\Psi^K: \mathcal{K}(E) \rightarrow \mathcal{K}(E/K)$ passes, by Proposition 3.9, to the localizations over $U_E(H)$ and $U_{E/K}(H/K)$, respectively. On the $\mathbb{Z}$ -graded endomorphisms rings, we get a homomorphism $\Psi^K: \mathcal{O}_E^\bullet(H) \rightarrow \mathcal{O}_{E/K}^\bullet(H/K)$ that on generators a_N, b_N is given by the formulas of Proposition 3.9. By Remark 5.8 this homomorphism $\Psi^K: \mathcal{O}_E^\bullet(H) \rightarrow \mathcal{O}_{E/K}^\bullet(H/K)$ is surjective.

6.11. **Proposition.** *For every elementary abelian group E , the spectrum $\text{Spc}(\mathcal{K}(E))$ admits a unique generic point η_E , namely the one of the cohomological open V_E .*

Proof. We proceed by induction on the p -rank. Let us write $\eta_E = \mathcal{P}_E(1, \sqrt{0})$ for the generic point of V_E , corresponding to the ideal $\sqrt{0}$ of nilpotent elements in $H^\bullet(E; k)$. Similarly, for every $K \leq E$, let us write $\eta_E(K) = \mathcal{P}_E(K, \eta_{E/K})$ for the generic point of the stratum $V_E(K) \simeq V_{E/K}$. We need to prove that every point $\eta_E(K)$ belongs to the closure of $\eta_E = \eta_E(1)$ in $\text{Spc}(\mathcal{K}(E))$. It suffices to show this for every cyclic subgroup $H < E$, by an easy induction argument on the rank, using the fact that $\psi^H: \text{Spc}(\mathcal{K}(E/H)) \hookrightarrow \text{Spc}(\mathcal{K}(E))$ is closed. So let $H \leq E$ be cyclic.

Note that inflation $\text{Infl}_E^{E/H}: \mathcal{K}(E/H) \rightarrow \mathcal{K}(E)$ passes to the localization of the former with respect to all $b_{N/H}$ for all $N \in \mathcal{N}(E)$ containing H (which is just the derived category of E/H) and of the latter with respect to the corresponding b_N :

$$(6.12) \quad \text{Infl}_E^{E/H}: \text{D}_b(k(E/H)) \longrightarrow \mathcal{K}(E)[\{b_N \mid N \geq H\}^{-1}].$$

This being a central localization of a fully-faithful functor with respect to a multiplicative subset in the source, it remains fully-faithful. One can further localize both categories with respect to all non-nilpotent $f \in H^\bullet(E/H; k)$ in the source, to obtain a fully-faithful

$$(6.13) \quad \text{Infl}_E^{E/H}: \text{D}_b(k(E/H))[\{f \mid f \notin \sqrt{0}\}^{-1}] \longrightarrow \mathcal{L}$$

where $\mathcal{L}$ is obtained from $\mathcal{K}(E)$ by first inverting all b_N for $N \geq H$ as in (6.12) and then inverting all $\text{Infl}_E^{E/H}(f)$ for $f \in H^\bullet(E/H; k) \setminus \sqrt{0}$.

At the level of spectra, $\text{Spc}(\mathcal{L})$ is a subspace of $\text{Spc}(\mathcal{K}(E))$. By construction, it meets the closed subset $\text{Im}(\psi^H) \cong \text{Spc}(\mathcal{K}(E/H))$ of $\text{Spc}(\mathcal{K}(E))$ only at the image of the generic point $\eta_E(H)$. Indeed, inverting all b_N for $N \geq H$ on $\text{Im}(\psi^H)$

corresponds to inverting all $b_{N/H}$ in $\mathcal{K}(E/H)$, hence shows that $\mathrm{Spc}(\mathcal{L}) \cap \mathrm{Im}(\psi^H)$ is in the image under ψ^H of the cohomological open $V_{E/H}$. Similarly, inverting all $f \notin \sqrt{0}$ removes all non-generic points of $V_{E/H}$. In particular, the generic point $\eta_E(H)$ of $V_E(H)$ is now a closed point of the subspace $\mathrm{Spc}(\mathcal{L})$ of $\mathrm{Spc}(\mathcal{K}(E))$.

Using that (6.13) is fully-faithful and that the endomorphism ring of the source is the cohomology of E/H localized at its generic point (in particular not a product of two rings), we see that $\mathcal{L}$ is not a product of two tt-categories and therefore $\mathrm{Spc}(\mathcal{L})$ is not disconnected. Also η_E belongs to $\mathrm{Spc}(\mathcal{L})$ and is distinct from $\eta_E(H)$. Hence the closed point $\eta_E(H) \in \mathrm{Spc}(\mathcal{L})$ cannot be isolated. Thus $\eta_E(H)$ belongs to the closure of some other point in $\mathrm{Spc}(\mathcal{L})$.

Let then $\mathcal{Q} \in \mathrm{Spc}(\mathcal{K}(E))$ be a point in the subspace $\mathrm{Spc}(\mathcal{L})$, such that $\mathcal{Q} \neq \eta_E(H)$ and $\eta_E(H) \in \overline{\{\mathcal{Q}\}}$, which reads $\mathcal{Q} \subsetneq \eta_E(H)$. We know by [BG22, Corollary 6.13] that this can only occur for $\mathcal{Q} = \mathcal{P}(H', \mathfrak{p})$ with $H' \leq H$, that is, either $H' = H$ or $H' = 1$ since here H was taken cyclic. The case $H' = H$ is excluded, as in the subspace $\mathrm{Spc}(\mathcal{L})$ the only prime of the form $\mathcal{P}(H, \mathfrak{q})$ that remained was $\eta_E(H)$ itself, and $\mathcal{Q}$ is different from $\eta_E(H)$. Thus $H' = 1$, which means that $\mathcal{Q} \in V_E = \overline{\{\eta_E(1)\}}$ and we therefore have $\eta_E(H) \in \overline{\{\mathcal{Q}\}} \subseteq \overline{\{\eta_E(1)\}}$ as claimed. $\square$

We can now determine the Krull dimension of the spectrum of $\mathcal{K}(E)$.

6.14. Proposition. *Let E be an elementary abelian p -group. Then the Krull dimension of $\mathrm{Spc}(\mathcal{K}(E))$ is the p -rank of E .*

Proof. By Proposition 4.11, the dimension of $\mathrm{Spc}(\mathcal{K}(E))$ is the maximum of the dimensions of the open subsets $U(H)$, for $H \leq E$. Each of these spaces has the same generic point η_E (by Proposition 6.11) and a unique closed point $\mathcal{M}(H)$ by Proposition 4.11 (and the fact that $\mathcal{M}(K) \in U(H)$ forces K and H to be contained in the same subgroups $N \in \mathcal{N}(G)$ by Proposition 3.9, which in turn forces $K = H$ because E is elementary abelian). Using Theorem 6.3 we translate the problem into one about the graded ring $\mathcal{O}_E^\bullet(H)$. Let $\eta_E = \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathcal{M}(H)$ be a chain of homogeneous prime ideals in $\mathcal{O}_E^\bullet(H)$. Note that $\mathfrak{p}_{n-1}$ belongs to the open $Z(f)^c$ of $\mathrm{Spec}^h(\mathcal{O}_E^\bullet(H))$ for some $f = \zeta_N^+$, $H \leq N$, or some $f = \zeta_N^-$, $H \not\leq N$. Each of these has non-zero degree so the graded ring $\mathcal{O}_E^\bullet(H)[f^{-1}]$ is periodic. We deduce that $\dim(\mathrm{Spec}^h(\mathcal{O}_E^\bullet(H)))$ is the maximum of $1 + \dim(R)$ where R ranges over the ungraded rings $R = \mathcal{O}_E^\bullet(H)[f^{-1}]_{(0)}$ for f as above. The reduced ring R_{red} is a finitely generated k -algebra with irreducible spectrum, hence a domain. Therefore $\dim(R) = \dim(R_{\mathrm{red}})$ is the transcendence degree of the residue field at the unique generic point. As observed above, this generic point is the same for all $H \leq E$, namely the generic point of $U(1) = \mathrm{Spc}(\mathrm{D}_b(kE))$. We conclude that $\dim(\mathrm{Spc}(\mathcal{K}(E))) = \dim(\mathrm{Spc}(\mathrm{D}_b(kE)))$ which is indeed the p -rank of E . $\square$

6.15. Remark. In fact, the proof shows that all closed points $\mathcal{M}(H) \in \mathrm{Spc}(\mathcal{K}(E))$ have the same codimension (height), namely the p -rank of E .

6.16. Remark. Thus for E elementary abelian, the Krull dimension of $\mathrm{Spc}(\mathcal{K}(E))$ is the same as the Krull dimension of the classical cohomological open $\mathrm{Spc}(\mathrm{D}_b(kE)) \cong \mathrm{Spec}^h(H^\bullet(E, k))$. In other words, the spectrum of $\mathcal{K}(E)$ is not monstrously different from that of $\mathrm{D}_b(kE)$, at least in terms of dimension, or ‘vertical complexity’. There is however ‘horizontal complexity’ in $\mathrm{Spc}(\mathcal{K}(E))$: each $U(H)$ has its own shape and form, and there are as many $U(H)$ as there are subgroups $H \leq E$. We give a finite presentation of the corresponding k -algebras $\mathcal{O}_E^\bullet(H)$ in Section 8.

7. CLOSURE IN ELEMENTARY ABELIAN CASE

In this section, E is still an elementary abelian p -group. Following up on Remark 2.11, we can now use Theorem 6.3 to analyze inclusion of tt-primes $\mathcal{P}, \mathcal{Q}$ in $\mathcal{K}(E)$, which amounts to asking when $\mathcal{Q}$ belongs to $\overline{\{\mathcal{P}\}}$ in $\mathrm{Spc}(\mathcal{K}(E))$.

7.1. *Remark.* Using again that every $\psi^H: \mathrm{Spc}(\mathcal{K}(E/H)) \hookrightarrow \mathrm{Spc}(\mathcal{K}(E))$ is a closed immersion, induction on the p -rank easily reduces the above type of questions to the case where the ‘lower’ point $\mathcal{P}$ belongs to $U_E(1) = V_E$. More generally, given a closed piece Z of the cohomological open V_E , we consider its closure $\bar{Z}$ in $\mathrm{Spc}(\mathcal{K}(E)) = \sqcup_{H \leq E} V_E(H)$ and we want to describe the part $\bar{Z} \cap V_E(H)$ in each stratum $V_E(H) \cong V_{E/H}$ for $H \leq E$.

7.2. *Construction.* Let $H \leq E$ be a subgroup of our elementary abelian group E . Consider the open subsets $U_E(H)$ of Proposition 4.11, the cohomological open $U_E(1) = V_E$ and their intersection $U_E(H) \cap V_E$. Consider also the stratum $V_E(H) = \check{\psi}^H(V_{E/H})$, that is a closed subset of $U_E(H)$ homeomorphic to $V_{E/H}$ via $\check{\psi}^H$:

$$\begin{array}{ccccc} & & U_E(H) & & V_E \\ & \nearrow \check{\psi}^H & & \nwarrow & \nearrow \\ V_{E/H} & & & & U_E(H) \cap V_E \end{array}$$

On graded endomorphism rings of the unit (Definition 5.9) this corresponds to

$$(7.3) \quad \begin{array}{ccccc} & & \mathcal{O}_E^\bullet(H) & & H^\bullet(E; k) \\ & \nwarrow \Psi^H & & \swarrow Q & \nwarrow Q' \\ H^\bullet(E/H; k) & & & & \mathcal{O}_E^\bullet(U_E(H) \cap V_E) \end{array}$$

where Q is the localization of $\mathcal{O}_E^\bullet(H)$ with respect to $\zeta_N^- = \frac{b_N}{a_N}$ for all $N \not\geq H$, where Q' is the localization of $\mathcal{O}_E^\bullet(1) = H^\bullet(E, k)$ with respect to $\zeta_N^+ = \frac{a_N}{b_N}$ for all $N \not\geq H$ and where Ψ^H is the epimorphism of Remark 6.10 for $K = H$.

7.4. **Lemma.** *With above notation, let $I \subseteq H^\bullet(E, k)$ be a homogeneous ideal of the cohomology of E . Define the homogeneous ideal $J = \Psi^H(Q^{-1}(\langle Q'(I) \rangle))$ in the cohomology $H^\bullet(E/H; k)$ of E/H by ‘carrying around’ the ideal I along (7.3):*

$$\begin{array}{ccccc} & & Q^{-1}(\langle Q'(I) \rangle) & & I \\ & \nwarrow & & \swarrow & \nwarrow \\ J := \Psi^H(Q^{-1}(\langle Q'(I) \rangle)) & & & & \langle Q'(I) \rangle \end{array}$$

Let Z be the closed subset of V_E defined by the ideal I . Then the closed subset of $V_{E/H}$ defined by J is exactly the intersection $\bar{Z} \cap V_E(H)$ of the closure $\bar{Z}$ of Z in $\mathrm{Spc}(\mathcal{K}(E))$ with the subspace $V_{E/H}$, embedded via $\check{\psi}^H$.

Proof. Once translated by Theorem 6.3, it is a general result about the multi-graded ring $A = H^{\bullet\bullet}(E)$. We have two open subsets, $U(H) = \cap_{s \in S_H} Z(s)^c$ and $V_E = U(1) = \cap_{s \in S_1} Z(s)^c$ for the multiplicative subsets S_H and S_1 of Definition 5.9. These open subsets are ‘Dirac-affine’, meaning they correspond to the homogenous spectra of the $\mathbb{Z}$ -graded localizations $S_H^{-1}(A)_{0\text{-twist}} = \mathcal{O}_E^\bullet(H)$ and $S_1^{-1}(A)_{0\text{-twist}} =$

$\mathcal{O}_E^\bullet(1) = \mathbf{H}^\bullet(E; k)$, where $(-)_0\text{-twist}$ refers to ‘zero-twist’, as before. The intersection of those two affine opens corresponds to inverting both S_H and S_1 , that is, inverting $\{ \frac{b_N}{a_N} \mid N \not\geq H \}$ from $\mathcal{O}_E^\bullet(H)$ and $\{ \frac{a_N}{b_N} \mid N \not\geq H \}$ from $\mathbf{H}^\bullet(E; k)$. This explains the two localizations Q and Q' and why their targets coincide.

The intersection $U(H) \cap \bar{Z}$ coincides with the closure in $U(H)$ of $U(H) \cap Z$. The latter is a closed subset of $U(H) \cap V_E$ defined by the ideal $\langle Q'(I) \rangle$. The preimage ideal $Q^{-1}(\langle Q'(I) \rangle)$ then defines that closure $U(H) \cap \bar{Z}$ in $U(H)$. Finally, to further intersect this closed subset of $U(H)$ with the closed subset $V_{E/H} = \text{Im}(\text{Spec}^h(\Psi^H))$, it suffices to project the defining ideal along the corresponding epimorphism $\Psi^H: \mathcal{O}_E^\bullet(H) \twoheadrightarrow \mathbf{H}^\bullet(E/H; k)$. $\square$

Before illustrating this method, we need a technical detour via polynomials.

7.5. Lemma. *Let I be a homogeneous ideal of the cohomology $\mathbf{H}^\bullet(E, k)$ and let $1 \neq H \not\leq E$ be a fixed non-trivial subgroup. Suppose that the only homogeneous ideal containing I and all the elements ζ_N for $N \geq H$ (Remark 3.7) is the maximal ideal $\mathbf{H}^+(E, k)$. Then there exists in I a homogeneous⁽¹⁾ polynomial f of the form*

$$f = \prod_{M \not\geq H} \zeta_M^d + \sum_m \lambda_m \cdot \prod_{N \in \mathbb{N}} \zeta_N^{m(N)}$$

for some integer $d \geq 1$ and scalars $\lambda_m \in k$ and finitely many exponents $m \in \mathbb{N}^{\mathbb{N}}$ that satisfy the following properties:

$$(7.6) \quad m(N) \geq 1 \text{ for at least one } N \geq H \quad \text{and} \quad m(N') < d \text{ for all } N' \not\geq H.$$

Proof. For simplicity, we work in the subring $\mathbf{H}^* \subseteq \mathbf{H}^\bullet(E, k)$ generated by the ζ_N . (For $p = 2$, this is the whole cohomology anyway and for p odd we only miss nilpotent elements, which are mostly irrelevant for the problem, as we can always raise everything in sight to a large p -th power.) Let us denote the maximal ideal by $\mathfrak{m} = \langle \zeta_N \mid N \in \mathbb{N} \rangle$. It is also convenient to work in the quotient $\mathbb{N}$ -graded ring

$$A^* := \mathbf{H}^*(E, k)/I$$

which is generated, as a k -algebra, by the classes $\bar{\zeta}_N$ of all ζ_N modulo I .

The assumption about $Z(I + \{ \zeta_N \mid N \geq H \}) = \{ \mathfrak{m} \}$ implies that $\mathfrak{m}$ has some power contained in $I + \langle \zeta_N \mid N \geq H \rangle$. In other words when $N' \not\geq H$ we have

$$(7.7) \quad (\bar{\zeta}_{N'})^d \in \langle \bar{\zeta}_N \mid N \geq H \rangle_{A^*}$$

for $d \gg 1$ that we take large enough to work for all the (finitely many) $N' \not\geq H$.

Consider this ideal $J = \langle \bar{\zeta}_N \mid N \geq H \rangle$ of A^* more carefully. It is a k -linear subspace generated by the classes $\bar{\theta}_m$ of the following products in $\mathbf{H}^*$

$$(7.8) \quad \theta_m := \prod_{N \in \mathbb{N}} (\zeta_N)^{m(N)}$$

with $m \in \mathbb{N}^{\mathbb{N}}$ such that $m(N) \geq 1$ for at least one $N \geq H$. We claim that J is in fact generated over k by the subset of the θ_m for the special $m \in \mathbb{N}^{\mathbb{N}}$ satisfying (7.6). Indeed, let $J' \subseteq J$ be the k -subspace generated by the θ_m for the special m . Then we can prove that the class $\bar{\theta}_m$ of each product (7.8) belongs to J' , by using (7.7) and (descending) induction on the number $\sum_{N \geq H} m(N)$. We conclude that $J = J'$.

¹ The grading is the usual $\mathbb{N}$ -grading in which all the ζ_N have the same degree $2'$. In particular, the first term $\prod_{M \not\geq H} \zeta_M^d$ in f has degree $2' \cdot d \cdot |\{ M \in \mathbb{N} \mid M \not\geq H \}|$.

By (7.7), the monomial $\prod_{M \not\geq H} (\bar{\zeta}_M)^d$ belongs to J and therefore to J' : It is a k -linear combination of monomials of the form $\bar{\theta}_m$ for $m \in \mathbb{N}^{\mathbb{N}}$ satisfying (7.6). Returning from $A^* = H^*(E, k)/I$ to $H^*(E, k)$, the difference between $\prod_{M \not\geq H} (\zeta_M)^d$ and the same k -linear combination of the lifts θ_m in $H^*(E, k)$ is an element of I , that we call f and that fulfills the statement. $\square$

7.9. Proposition. *Let $Z \subset V_E$ be a non-empty closed subset of the cohomological open and let $1 \neq H \leq E$ be a non-trivial subgroup. Suppose that in V_E , the subset Z intersects the image of the cohomological open of H in the smallest possible way:*

$$Z \cap \rho_H(V_H) = \{\mathcal{M}(1)\}.$$

Consider the closure $\bar{Z}$ of Z in the whole spectrum $\mathrm{Spc}(\mathcal{K}(E))$. Then $\bar{Z}$ does not intersect the stratum $V_E(H) = \psi^H(V_{E/H})$. Hence $\mathcal{M}(H)$ does not belong to $\bar{Z}$.

Proof. Let $I \subset H^*(E, k)$ be the homogeneous ideal that defines Z . The closed image $\rho_H(V_H)$ is given by the (partly redundant) equations $\zeta_N = 0$ for all $N \geq H$. It follows that the intersection $Z \cap \rho_H(V_H)$ is defined by the ideal $I + \langle \zeta_N \mid N \geq H \rangle$. So our hypothesis translates exactly in saying that I satisfies the hypothesis of Lemma 7.5. Hence there exists a homogeneous element of I

$$f = \prod_{M \not\geq H} \zeta_M^d + \sum_m \lambda_m \prod_{N \in \mathbb{N}} \zeta_N^{m(N)}$$

for scalars $\lambda_m \in k$ and finitely many exponents $m \in \mathbb{N}^{\mathbb{N}}$ satisfying (7.6). We can now use Lemma 7.4 and follow Diagram (7.3) with the ideal I and particularly with its element f . The element $Q'(f)$ is just f seen in $\mathcal{O}_E^\bullet(U_E(H) \cap V_E)$. But it does not belong to the image of $\mathcal{O}_E^\bullet(H)$ under Q because f contains some b_M with $M \not\geq H$ in denominators in the ζ_M 's. Still, we can multiply $Q'(f)$ by $\prod_{M \not\geq H} (\frac{b_M}{a_M})^d = \prod_{M \not\geq H} (\zeta_M)^{-d}$ to get a degree-zero homogeneous element

$$(7.10) \quad \tilde{f} = 1 + \sum_m \lambda_m \prod_{N \in \mathbb{N}} \zeta_N^{m'(N)}$$

in the ideal $\langle Q'(f) \rangle$, where we set the exponent $m'(N) := m(N) - d$ if $N \not\geq H$ and $m'(N) := m(N)$ if $N \geq H$. Note that by (7.6) the exponent $m'(N)$ is negative if $N \not\geq H$ and is non-negative if $N \geq H$ and strictly positive for at least one $N \geq H$. Both types of exponents of ζ_N are allowed in $\mathcal{O}_E^\bullet(H)$, namely, when $N \not\geq H$, the element $\zeta_N^- = \frac{b_N}{a_N}$ exists in $\mathcal{O}_E^\bullet(H)$. In other words, the element $\tilde{f} \in \langle Q'(f) \rangle$ satisfies

$$\tilde{f} = Q(1 + \tilde{g})$$

where $\tilde{g} \in \mathcal{O}_E^\bullet(H)$ belongs to the ideal $\langle \zeta_N \mid N \geq H \rangle$ in $\mathcal{O}_E^\bullet(H)$ and must be of degree zero by homogeneity. Now, for $N \geq H$, we have $\Psi^H(\zeta_N) = \zeta_{N/H}$ by Proposition 3.9. It follows that $\Psi^H(\tilde{g})$ belongs to the maximal ideal $\langle \zeta_{\bar{N}} \mid \bar{N} \in \mathcal{N}(E/H) \rangle \subseteq H^+(E/H, k)$ of $H^*(E/H, k)$ and still has degree zero. This forces $\Psi^H(\tilde{g}) = 0$ and therefore $\Psi^H(1 + \tilde{g}) = 1$ in $H^*(E/H, k)$. In the notation of Lemma 7.4, we have shown that J contains 1, which implies that $\bar{Z} \cap V_E(H) = \emptyset$. $\square$

7.11. Corollary. *Let $Z \subset V_E$ be a closed subset of the cohomological open, strictly larger than the unique closed point $\mathcal{M}(1)$ of V_E . Suppose that in V_E , the subset Z intersects the images of all proper subgroups trivially, i.e. $Z \cap (\bigcup_{H \leq E} \rho_H(V_H)) = \{\mathcal{M}(1)\}$. Then the closure $\bar{Z}$ of Z in the whole spectrum $\mathrm{Spc}(\mathcal{K}(E))$ has only one more point, namely $\bar{Z} = Z \cup \{\mathcal{M}(E)\}$.*

Proof. By Proposition 7.9, we see that $\bar{Z}$ does not meet any stratum $V_E(H)$ for $H \neq E$. Thus the only point of $\mathrm{Spc}(\mathcal{K}(E))$ outside of Z itself, hence outside of V_E , that remains candidate to belong to $\bar{Z}$ must belong to $\mathrm{supp}(\mathcal{K}_{\mathrm{ac}}(E)) \setminus \cup_{H \not\leq E} V_E(H) = V_E(E) = \{\mathcal{M}(E)\}$. We know that $\mathcal{M}(E) = \langle k(E/H) \mid H \not\leq E \rangle$ in $\mathcal{K}(E)$, by [BG22, Example 6.30]. Take $\mathcal{P} \in Z$ different from $\mathcal{M}(1)$. Since $\mathcal{P}$ does not belong to any $\mathrm{Im}(\rho_H) = \mathrm{supp}(k(E/H))$ by assumption, it must contain $k(E/H)$. Consequently, $\mathcal{M}(E) \subseteq \mathcal{P}$, meaning that $\mathcal{M}(E) \in \{\mathcal{P}\} \subseteq \bar{Z}$. $\square$

7.12. Corollary. *Let E be an elementary abelian group of rank r . Let $\mathcal{P}$ be a point of height $r-1$ in the cohomological open V_E , that is, a closed point of the classical projective support variety $\mathcal{V}_E(k) := V_E \setminus \{\mathcal{M}(1)\} \cong \mathrm{Proj}(\mathrm{H}^*(E, k)) \cong \mathbb{P}_k^{r-1}$. Suppose that $\mathcal{P}$ does not belong to the image $\rho_H(V_H)$ of the support variety of any proper subgroups $H \not\leq E$. Then the closure of $\mathcal{P}$ in $\mathrm{Spc}(\mathcal{K}(E))$ is exactly the following*

$$\overline{\{\mathcal{P}\}} = \{\mathcal{M}(E), \mathcal{P}, \mathcal{M}(1)\}.$$

Proof. Apply Corollary 7.11 to $Z = \{\mathcal{P}, \mathcal{M}(1)\}$, the closure of $\mathcal{P}$ in V_E . $\square$

7.13. Example. We can review the proof of Proposition 7.9 in the special case of Corollary 7.12, to see how elements like $f \in \mathcal{O}_E^\bullet(1)$ and $\tilde{f} \in \mathcal{O}_E^\bullet(H)$ come into play. We do it in the special case where $\mathcal{P}$ is a k -rational point (e.g. if k is algebraically closed). Let $1 \neq H \not\leq E$ be a non-trivial subgroup (the case $r = 1$ being trivial). Choose $N_0, N_1 \triangleleft E$ index- p subgroups with $H \leq N_0$ and $H \not\leq N_1$. They define coordinates ζ_0, ζ_1 in $\mathbb{P}^{r-1}$ (where $\zeta_i = \zeta_{N_i}$ as in Remark 3.7). There exists a hyperplane of $\mathbb{P}^{r-1}$

$$(7.14) \quad \lambda_0 \zeta_0 + \lambda_1 \zeta_1 = 0, \quad [\lambda_0 : \lambda_1] \in \mathbb{P}^1(k),$$

going through the rational point $\mathcal{P}$. Note that $\lambda_1 \neq 0$ as $\mathcal{P} \notin Z(\zeta_0) = \rho_{N_0}(V_{N_0})$, by assumption. As in Lemma 7.4, the following two localizations agree

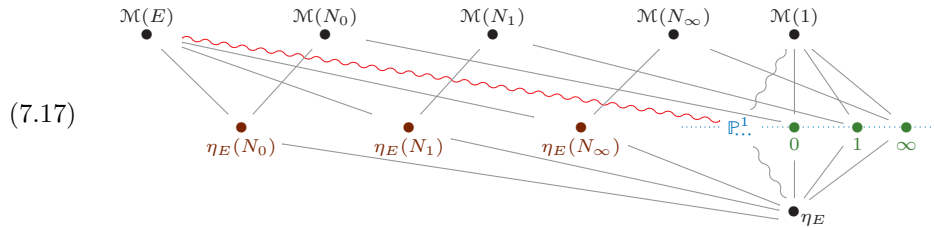
$$(7.15) \quad \mathcal{O}_E^\bullet(H)[(\zeta_N^-)^{-1} \mid H \not\leq N] = \mathcal{O}_E^\bullet(1)[(\zeta_N^+)^{-1} \mid H \not\leq N]$$

where $N \triangleleft E$ ranges over the index- p subgroups as usual. We find a lift

$$\tilde{f} := \lambda_0 \zeta_0 \zeta_{N_1}^- + \lambda_1 \in \mathcal{O}_E^\bullet(H)$$

of the element $f = \lambda_0 \zeta_0 + \lambda_1 \zeta_1 \in \mathcal{O}_E^\bullet(1)$ of (7.14) suitably multiplied by $\zeta_1^{-1} = \zeta_{N_1}^-$ in the localization (7.15). Then we have $\Psi^H(\zeta_{N_1}^-) = 0$ since $H \not\leq N_1$, by Proposition 3.9, so $\Psi^H(\tilde{f}) = \lambda_1 \in k^\times$ is an isomorphism. We deduce that $\mathrm{cone}(\tilde{f})$ belongs to $\mathcal{M}(H) \setminus \mathcal{P}$, which shows that $\mathcal{P}$ does not specialize to $\mathcal{M}(H)$.

7.16. Example. Let $E = C_2 \times C_2$ be the Klein-four group. Let us justify the description of $\mathrm{Spc}(\mathcal{K}(E))$ announced in [BG22, Example 7.10] in some detail:



By Recollection 1.8, we have a partition of the spectrum as a set

$$\mathrm{Spc}(\mathcal{K}(E)) = V_E(E) \sqcup V_E(N_0) \sqcup V_E(N_1) \sqcup V_E(N_\infty) \sqcup V_E,$$

where we write N_0, N_1, N_∞ for the three cyclic subgroups C_2 and where $V_E = V_E(1)$ is the cohomological open as usual. Let us review those five parts $V_E(H) = \psi^H(V_{E/H})$ separately, in growing order of complexity, i.e. from left to right in (7.17).

For $H = E$, the stratum $V_E(E) = \psi^E(V_{E/E}) = \{\mathcal{M}(E)\}$ is just a closed point.

For each cyclic subgroup $N_i < E$, the quotient $E/N_i \simeq C_2$ is cyclic, so $\mathrm{Spc}(E/N_i)$ is the space of Example 6.8. Its image under ψ^{N_i} is $\{\mathcal{M}_E(E), \eta_E(N_i), \mathcal{M}_E(N_i)\}$, defining the (brown) point $\eta_E(N_i) := \psi^{N_i}(\eta_{E/N_i})$, as in the proof of Proposition 6.11. The stratum $V_E(N_i)$ is the image of the cohomological open V_{E/N_i} only, that is, the Sierpiński space $\{\eta_E(N_i), \mathcal{M}(N_i)\}$, whose non-closed point $\eta_E(N_i)$ is the generic point of the irreducible $\{\mathcal{M}_E(E), \eta_E(N_i), \mathcal{M}_E(N_i)\}$ in $\mathrm{Spc}(\mathcal{K}(E))$.

Finally, for $H = 1$, the cohomological open $V_E = \mathrm{Spc}(\mathrm{D}_b(kE)) \cong \mathrm{Spec}^h(k[\zeta_0, \zeta_1])$ is a $\mathbb{P}_k^1$ with a closed point $\mathcal{M}(1)$ on top. We denote by η_E the generic point of $\mathrm{Spc}(\mathcal{K}(E))$ as in Proposition 6.11 and by $0, 1, \infty$ the three $\mathbb{F}_2$ -rational points of $\mathbb{P}_k^1$ (in green). The notation $\mathbb{P}_k^1$ refers to all remaining points of $\mathbb{P}_k^1$. The undulated lines indicate that *all* points of $\mathbb{P}_k^1$ have the same behavior. Namely, η_E specializes to all points of $\mathbb{P}_k^1$ and every point of $\mathbb{P}_k^1$ specializes to $\mathcal{M}(1)$ and the (red) undulated line towards $\mathcal{M}(E)$ indicates that all points of $\mathbb{P}_k^1$ specialize to $\mathcal{M}(E)$, as follows from Corollary 7.12. (Note that the latter was rather involved: Its proof occupies most of this section, and relies on technical Lemma 7.5.)

We have described the closure of every point in $\mathrm{Spc}(\mathcal{K}(E))$, except for the $\mathbb{F}_2$ -rational points $0, 1, \infty$. For this, we use the closed immersion $\rho_{N_i}: \mathrm{Spc}(\mathcal{K}(N_i)) \hookrightarrow \mathrm{Spc}(\mathcal{K}(E))$ induced by restriction Res_{N_i} . The point i is the image of the generic point η_{N_i} of the V-shaped space $\mathrm{Spc}(\mathcal{K}(N_i))$ of Example 6.8. Hence its closure is $\mathrm{Im}(\rho_{N_i}) = \{\mathcal{M}(E_i), i, \mathcal{M}(1)\}$. So specializations are exactly those of (7.17).

We revisit this picture in more geometric terms in Example 9.2.

7.18. Remark. It is possible to extend Corollary 7.12 to a general finite group G by means of the Colimit Theorem 2.10. Let $Z \subseteq \mathrm{Spc}(\mathcal{K}(G))$ be a one-dimensional irreducible closed subset. Write its generic point as $\mathcal{P} = \mathcal{P}(K, \mathfrak{p})$ for (unique) $K \in \mathrm{Sub}_p(G)/G$ and $\mathfrak{p} \in V_{G//K}$. By Quillen applied to $\bar{G} = G//K$, there exists a minimal elementary abelian subgroup $E \leq \bar{G}$ such that $\mathfrak{p} \in \mathrm{Im}(\rho_E: V_E \rightarrow V_{\bar{G}})$, also unique up to $\bar{G}$ -conjugation. This $E \leq \bar{G} = (N_G K)/K$ is given by $E = H/K$ for $H \leq N_G K$ containing K . Then $\mathcal{P} = \varphi_{(H,K)}(\mathcal{Q})$ where $\mathcal{Q} \in \mathrm{Spc}(\mathcal{K}(E))$ is given by $\mathcal{Q} = \mathcal{P}_E(1, \mathfrak{q})$ for some $\mathfrak{q} \in V_E$. By Lemma 2.9, the map $\varphi_{(H,K)}: \mathrm{Spc}(\mathcal{K}(E)) \rightarrow \mathrm{Spc}(\mathcal{K}(G))$ is closed and preserves the dimension of points. It follows that $\mathcal{Q}$ is also the generic point of a one-dimensional irreducible in $\mathrm{Spc}(\mathcal{K}(E))$. By minimality of E , the point $\mathcal{Q} \in V_E$ does not belong to $V_{H'}$ for any proper subgroup $H' < E$. By Corollary 7.12, we have $\overline{\{\mathcal{Q}\}} = \{\mathcal{M}_E(E), \mathcal{Q}, \mathcal{M}_E(1)\}$ in $\mathrm{Spc}(\mathcal{K}(E))$. The map $\varphi_{(H,K)}$ sends this subset to $\{\mathcal{M}_G(H), \mathcal{P}, \mathcal{M}_G(K)\}$. In summary, every one-dimensional irreducible subset of $\mathrm{Spc}(\mathcal{K}(G))$ is of the form $Z = \{\mathcal{M}(H), \mathcal{P}, \mathcal{M}(K)\}$, where H and K are uniquely determined by the generic point $\mathcal{P}$ via the above method.

8. PRESENTATION OF TWISTED COHOMOLOGY

We remain in the case of an elementary abelian group E . In this section we want to better understand the local $\mathbb{Z}$ -graded rings $\mathcal{O}_E^\bullet(H)$ that played such an important role in Section 6. Thankfully they are reasonable k -algebras.

8.1. Terminology. Recall that we write $C_p = \langle \sigma \mid \sigma^p = 1 \rangle$ for the cyclic group of order p with a chosen generator σ . For brevity we call an $\mathbb{F}_p$ -linear surjection

$\pi: E \rightarrow C_p$ a *coordinate*. For two coordinates π, π' we write $\pi \sim \pi'$ if $\ker(\pi) = \ker(\pi')$. Finally, for a subgroup H , we often abbreviate $H \mid \pi$ to mean $H \leq \ker(\pi)$.

Recall from Definition 3.3 and Remark 3.5 that each coordinate π yields an invertible object $u_\pi = \pi^* u_p$ in $\mathcal{K}(E)$. It comes with maps $a_\pi, b_\pi, c_\pi: k \rightarrow u_\pi[*]$.

8.2. Remark. If $\pi \sim \pi'$ then there exists a unique $\lambda \in \mathbb{F}_p^\times$ such that $\pi' = \pi^\lambda$. Hence, if $p = 2$ then necessarily $\pi = \pi'$ and $u_\pi = u_{\pi'}$. On the other hand, if $p > 2$ is odd then we still have $u_\pi \cong u_{\pi'}$ as already mentioned. Explicitly, consider the automorphism $\lambda: C_p \rightarrow C_p$ that sends σ to σ^λ . The isomorphism $u_\pi = \pi^* u_p \xrightarrow{\sim} \pi^* \lambda^* u_p = (\pi^\lambda)^* u_p = u_{\pi'}$ will be the pullback $\pi^* \Lambda$ along π of an isomorphism of complexes $\Lambda: u_p \xrightarrow{\sim} \lambda^* u_p$. This isomorphism Λ can be given explicitly by the identity in degree 0 and by the kC_p -linear maps $kC_p \rightarrow \lambda^* kC_p$ in degree 1 (resp. 2) determined by $1 \mapsto 1$ (resp. $1 \mapsto 1 + \sigma + \cdots \sigma^{\lambda-1}$). One checks directly that $\Lambda \circ a_p = a_p$ and $\Lambda \circ b_p = \lambda \cdot b_p$. By applying π^* we obtain

$$(8.3) \quad (\pi^* \Lambda) \circ a_\pi = a_{\lambda\pi} \quad \text{and} \quad (\pi^* \Lambda) \circ b_\pi = \lambda \cdot b_{\lambda\pi}.$$

8.4. Lemma. *Given coordinates $\pi_1 \not\sim \pi_2$ set $\pi_3 = \pi_1^{-1} \pi_2^{-1}$. Write u_i, a_i and b_i for u_{π_i}, a_{π_i} and b_{π_i} in $\mathcal{K}(E)$. Then we have the relation*

$$a_1 b_2 b_3 + b_1 a_2 b_3 + b_1 b_2 a_3 = 0$$

as a map from $\mathbb{1}$ to $(u_1 \otimes u_2 \otimes u_3)[-2' \cdot 2]$ in $\mathcal{K}(E)$. (See Notation 3.2 for $2'$.)

Proof. Let $N_i = \ker(\pi_i)$ for $i = 1, 2, 3$, which are all distinct. Let $N = N_1 \cap N_2 \cap N_3$ be the common kernel, which is of index p^2 in E . By inflation along $E \rightarrow E/N$, it suffices to prove the lemma for $E = C_p \times C_p$ and π_1 and π_2 the two projections on the factors. We abbreviate u for the complex of permutation kE -modules $u := u_1 \otimes u_2 \otimes u_3$. Consider the permutation module $M := kC_p \otimes kC_p \otimes kC_p \cong k(E/N_1) \otimes k(E/N_2) \otimes k(E/N_3)$ which appears as a summand in various degrees of the complex u . One element in M is of particular interest:

$$m := \sum_{i_1, i_2=0}^{p-1} \sigma^{i_1} \otimes \sigma^{i_2} \otimes \sigma^{-i_1-i_2}.$$

It is easy to check that m is E -invariant, thus defines a kE -linear map $\tilde{m}: k \rightarrow M$, that can be used to define the required homotopies. This depends on p . If $p = 2$, the homotopy is given by $\tilde{m}$ when viewed from $\mathbb{1}$ to the only M -entry of $u[-2]$ in degree one. If $p > 2$, the homotopy is given by $(\tilde{m}, \tilde{m}, \tilde{m})$ as a map from $\mathbb{1}$ to the three M -entries of $u[-4]$ in degree one. Verifications are left to the reader. $\square$

8.5. Construction. We construct a commutative k -algebra $\underline{\mathcal{Q}}_E^\bullet(H)$ by generators and relations. Its generators are indexed by coordinates $\pi: E \rightarrow C_p$ (Terminology 8.1)

$$\{ \zeta_\pi^+ \mid \pi \text{ s.t. } H \leq \ker(\pi) \} \cup \{ \zeta_\pi^- \mid \pi \text{ s.t. } H \not\leq \ker(\pi) \}.$$

These generators come equipped with a degree in $\mathbb{Z}$: If $H \mid \pi$ the generator ζ_π^+ is set to have degree $2'$, whereas if $H \nmid \pi$ the generator ζ_π^- is set to have degree $-2'$. We impose the following four families of homogeneous relations. First for every coordinate π and every $\lambda \in \mathbb{F}_p^\times$ (for p odd), we have a rescaling relation

$$(a) \quad \zeta_{\pi^\lambda}^+ = \lambda \zeta_\pi^+ \text{ if } H \mid \pi \quad \text{and} \quad \zeta_{\pi^\lambda}^- = \lambda^{-1} \zeta_\pi^- \text{ if } H \nmid \pi'$$

and whenever $\pi_3 = \pi_1^{-1} \pi_2^{-1}$ and $\pi_1 \not\sim \pi_2$, writing $\zeta_i^\pm := \zeta_{\pi_i}^\pm$, we impose one of the following relations, inspired by Lemma 8.4:

(b) $\zeta_1^+ + \zeta_2^+ + \zeta_3^+ = 0$, if $H \mid \pi_1$ and $H \mid \pi_2$ (and therefore $H \mid \pi_3$)

(c) $\zeta_1^- + \zeta_2^- + \zeta_1^- \zeta_2^- \zeta_3^+ = 0$, if $H \nmid \pi_1$ and $H \nmid \pi_2$ but $H \mid \pi_3$

(d) $\zeta_1^- \zeta_2^- + \zeta_2^- \zeta_3^- + \zeta_3^- \zeta_1^- = 0$ if $H \nmid \pi_i$ for all $i = 1, 2, 3$.

Since these relations are homogeneous, the ring $\underline{\mathcal{O}}_E^\bullet(H)$ is a $\mathbb{Z}$ -graded ring.

8.6. *Remark.* We could also define a multi-graded commutative k -algebra $\underline{H}^{\bullet\bullet}(E)$ generated by all a_π, b_π subject to the relations in (8.3) and Lemma 8.4. This algebra $\underline{H}^{\bullet\bullet}(E)$ would be $\mathbb{Z} \times \mathbb{N}^{\mathcal{N}}$ -graded with a_π in degree $(0, 1_{\ker(\pi)})$ and b_π in degree $(-2', 1_{\ker(\pi)})$. Then $\underline{\mathcal{O}}_E^\bullet(H)$ is simply the ‘zero-twist’ part of the localization of $\underline{H}^{\bullet\bullet}(E)$ with respect to the a_π, b_π that become invertible in $U(H)$, that is, those a_π such that $H \nmid \pi$ and those b_π such that $H \mid \pi$, following the pattern of Definition 5.9.

8.7. *Remark.* By (8.3) and Lemma 8.4, there exists a canonical homomorphism

$$(8.8) \quad \underline{\mathcal{O}}_E^\bullet(H) \rightarrow \mathcal{O}_E^\bullet(H)$$

mapping ζ_π^+ to $\frac{a_\pi}{b_\pi}$ and ζ_π^- to $\frac{b_\pi}{a_\pi}$.

8.9. *Example.* Let $H = 1$. Recall from Example 5.13 that $\mathcal{O}_E^\bullet(1)$ is the cohomology ring. Then the homomorphism (8.8) is the standard one $\underline{\mathcal{O}}_E^\bullet(1) \rightarrow H^\bullet(E; k)$, that maps ζ_π^+ to the usual generator $\zeta_\pi = \pi^*(\zeta_{C_p})$. Note that here $H = 1 \mid \pi$ for all π , so there is no ζ_π^- . For E elementary abelian, it is well-known that this homomorphism $\underline{\mathcal{O}}_E^\bullet(1) \rightarrow H^\bullet(E; k)$ is an isomorphism modulo nilpotents. See for instance [Car96].

For two subgroups $H, K \leq E$, the open subsets $U_E(H)$ and $U_E(K)$ can intersect in $\text{Spc}(\mathcal{K}(E))$. Similarly, we can discuss what happens with the rings $\underline{\mathcal{O}}_E^\bullet(H)$.

8.10. **Proposition.** *Let $H, K \leq E$ be two subgroups. Define $S = S(H, K) \subset \underline{\mathcal{O}}_E^\bullet(H)$ to be the multiplicative subset generated by the finite set*

$$\{\zeta_\pi^+ \mid \text{for } \pi \text{ with } H \mid \pi \text{ and } K \nmid \pi\} \cup \{\zeta_\pi^- \mid \text{for } \pi \text{ with } H \nmid \pi \text{ and } K \mid \pi\}$$

and similarly, swapping H and K , let $T = S(K, H) \subset \underline{\mathcal{O}}_E^\bullet(K)$ be the multiplicative subset generated by $\{\zeta_\pi^+ \mid H \nmid \pi \text{ and } K \mid \pi\} \cup \{\zeta_\pi^- \mid H \mid \pi \text{ and } K \nmid \pi\}$. Then we have a canonical isomorphism of (periodic) $\mathbb{Z}$ -graded rings

$$S^{-1}\underline{\mathcal{O}}_E^\bullet(H) \cong T^{-1}\underline{\mathcal{O}}_E^\bullet(K)$$

and in particular of their degree-zero parts. Thus the open of $\text{Spec}^h(\underline{\mathcal{O}}_E^\bullet(H))$ defined by S is canonically homeomorphic to the open of $\text{Spec}^h(\underline{\mathcal{O}}_E^\bullet(K))$ defined by T .

Proof. The left-hand side $S^{-1}\underline{\mathcal{O}}_E^\bullet(H)$ is the (‘zero-twist’ part of the) localization of the multi-graded ring $\underline{H}^{\bullet\bullet}(E)$ of Remark 8.6 with respect to

$$\begin{aligned} & \{a_\pi \mid H \nmid \pi\} \cup \{b_\pi \mid H \mid \pi\} \cup \{a_\pi \mid H \mid \pi, K \nmid \pi\} \cup \{b_\pi \mid H \nmid \pi, K \mid \pi\} \\ &= \{a_\pi \mid H \nmid \pi \text{ or } K \nmid \pi\} \cup \{b_\pi \mid H \mid \pi \text{ or } K \mid \pi\} \end{aligned}$$

which is symmetric in H and K . This completes the proof. $\square$

8.11. *Remark.* The above isomorphism is compatible with the homomorphism (8.8), namely the obvious diagram commutes when we perform the corresponding localizations on $\underline{\mathcal{O}}_E^\bullet(H)$ and $\underline{\mathcal{O}}_E^\bullet(K)$.

8.12. Proposition. *Let $K \leq H \leq E$. There is a canonical split epimorphism $\Psi^K: \underline{\mathcal{O}}_E^\bullet(H) \rightarrow \underline{\mathcal{O}}_{E/K}^\bullet(H/K)$ whose kernel is $\langle \zeta_\pi^- \mid K \nmid \pi \rangle$. It is compatible with the homomorphism Ψ^K of Remark 6.10, in that the following diagram commutes*

$$\begin{array}{ccc} \underline{\mathcal{O}}_E^\bullet(H) & \xrightarrow{(8.8)} & \mathcal{O}_E^\bullet(H) \\ \Psi^K \downarrow & & \downarrow \Psi^K \\ \underline{\mathcal{O}}_{E/K}^\bullet(H/K) & \xrightarrow{(8.8)} & \mathcal{O}_{E/K}^\bullet(H/K). \end{array}$$

Proof. Set $\bar{H} := H/K \leq \bar{E} := E/K$. Similarly, for every coordinate $\pi: E \rightarrow C_p$ such that $K \mid \pi$, let us write $\bar{\pi}: \bar{E} \rightarrow C_p$ for the induced coordinate. The morphism Ψ^K will come from a morphism $\Psi^K: \underline{\mathbf{H}}^{\bullet\bullet}(E) \rightarrow \underline{\mathbf{H}}^{\bullet\bullet}(E/K)$, with respect to the homomorphism of gradings (5.3). As these algebras are constructed by generators and relations (Remark 8.6), we need to give the image of generators. Inspired by Proposition 3.9 we define $\Psi^K: \underline{\mathbf{H}}^{\bullet\bullet}(E) \rightarrow \underline{\mathbf{H}}^{\bullet\bullet}(E/K)$ on generators by

$$a_\pi \mapsto \begin{cases} a_{\bar{\pi}} & \text{if } K \mid \pi \\ 1 & \text{if } K \nmid \pi \end{cases} \quad b_\pi \mapsto \begin{cases} b_{\bar{\pi}} & \text{if } K \mid \pi \\ 0 & \text{if } K \nmid \pi. \end{cases}$$

It is easy to see that the relations in $\underline{\mathbf{H}}^{\bullet\bullet}(E)$ are preserved; thus the map Ψ^K is well-defined. Let $\varpi: E \rightarrow E/K$ and for every $\bar{\pi}: \bar{E} \rightarrow C_p$ consider the coordinate $\pi = \bar{\pi} \circ \varpi: E \rightarrow C_p$. Then $\bar{H} \mid \bar{\pi}$ if and only if $H \mid \pi$. It follows that the morphism passes to the localizations $\Psi^K: \underline{\mathcal{O}}_E^\bullet(H) \rightarrow \underline{\mathcal{O}}_{E/K}^\bullet(H/K)$ as announced. The statement about its kernel is easy and commutativity of the square follows from the fact (Remark 6.10) that Ψ^K treats the a_π and b_π according to the same formulas.

The section of Ψ^K is inspired by inflation. Namely, $a_{\bar{\pi}} \mapsto a_\pi$ and $b_{\bar{\pi}} \mapsto b_\pi$ defines a map of graded k -algebras $\underline{\mathbf{H}}^{\bullet\bullet}(\bar{E}) \rightarrow \underline{\mathbf{H}}^{\bullet\bullet}(E)$ that is already a section to Ψ^K and passes to the localizations. $\square$

8.13. Theorem. *The canonical homomorphism (8.8) induces an isomorphism*

$$\underline{\mathcal{O}}_E^\bullet(H)_{\text{red}} \xrightarrow{\sim} \mathcal{O}_E^\bullet(H)_{\text{red}}$$

of reduced $\mathbb{Z}$ -graded k -algebras.

Proof. It follows from Remark 5.11 that the map is surjective. We will now show that the closed immersion $\text{Spec}^h(\mathcal{O}_E^\bullet(H)) \hookrightarrow \text{Spec}^h(\underline{\mathcal{O}}_E^\bullet(H))$ is surjective—this will complete the proof, by the usual commutative algebra argument, which can be found in [HP23, Lemma 2.22] for the graded case. By Theorem 6.3, this is equivalent to showing the surjectivity of the composite with comp_E , that we baptize β^H

$$(8.14) \quad \beta^H: U_E(H) \xrightarrow[\text{comp}_E]{\sim} \text{Spec}^h(\mathcal{O}_E^\bullet(H)) \hookrightarrow \text{Spec}^h(\underline{\mathcal{O}}_E^\bullet(H)).$$

We proceed by induction on the order of the subgroup H . If $H = 1$ the result follows from Example 8.9. So suppose that $H \neq 1$ and pick a homogeneous prime $\mathfrak{p} \in \text{Spec}^h(\underline{\mathcal{O}}_E^\bullet(H))$. We distinguish two cases.

If for every coordinate $\pi: E \rightarrow C_p$ such that $H \nmid \pi$ we have $\zeta_\pi^- \in \mathfrak{p}$ then $\mathfrak{p}$ belongs to $V(\langle \zeta_\pi^- \mid H \nmid \pi \rangle)$, which we identify with the image of $\text{Spec}^h(\underline{\mathcal{O}}_{E/H}^\bullet(1))$

by Proposition 8.12 applied to $K = H$. Namely, we have a commutative square

$$\begin{array}{ccc} U_E(H) & \xleftarrow{\psi^H} & U_{E/H}(1) \\ \beta^H \downarrow & & \downarrow \beta^1 \\ \mathrm{Spec}^h(\mathcal{Q}_E^\bullet(H)) & \xleftarrow{\mathrm{Spec}^h(\psi^H)} & \mathrm{Spec}^h(\mathcal{Q}_{E/H}^\bullet(1)) \end{array}$$

and since the right-hand vertical arrow is surjective by the case already discussed, we conclude that $\mathfrak{p}$ belongs to the image of β^H in (8.14) as well.

Otherwise, there exists a coordinate π_1 such that $H \nmid \pi_1$ and $\zeta_{\pi_1}^- \notin \mathfrak{p}$. Let $K := H \cap \ker(\pi_1)$ and let $S = S(H, K)$ be defined as in Proposition 8.10:

$$S = \{ \zeta_\pi^- \mid \text{for } \pi \text{ with } H \nmid \pi \text{ and } K \mid \pi \}.$$

We claim that $\mathfrak{p}$ belongs to the open of $\mathrm{Spec}^h(\mathcal{Q}_E^\bullet(H))$ defined by S . Indeed, let $\zeta_{\pi_2}^- \in S$, that is for π_2 with $H \nmid \pi_2$ and $K \mid \pi_2$, and let us show that $\zeta_{\pi_2}^- \notin \mathfrak{p}$. If $\pi_2 \sim \pi_1$ this is clear from $\zeta_{\pi_1}^- \notin \mathfrak{p}$ and the relation (a) in $\mathcal{Q}_E^\bullet(H)$. If $\pi_2 \not\sim \pi_1$, let $h \in H \setminus K$ (so that h generates the cyclic group $H/K \cong C_p$). As $\pi_1(h) \neq 1$ and $\pi_2(h) \neq 1$ we may replace π_1 by an equivalent coordinate $\tilde{\pi}_1 := \pi_1^\lambda$ such that $\tilde{\pi}_1(h) = \pi_2(h)^{-1}$ and therefore $H \mid \pi_3 := \tilde{\pi}_1^{-1}\pi_2^{-1}$. Then relation (c) exhibits $\zeta_{\tilde{\pi}_1}^-$ as a multiple of $\zeta_{\pi_2}^-$. As the former does not belong to $\mathfrak{p}$ (by the previous case), neither does $\zeta_{\pi_2}^-$. At this point we may apply Proposition 8.10 for our subgroups H and K . By Remark 8.11, we have a commutative triangle:

$$\begin{array}{ccc} & U_E(H) \cap U_E(K) & \\ \beta^H \swarrow & & \searrow \beta^K \\ \mathrm{Spec}^h(\mathcal{Q}_E^\bullet(H)[S^{-1}]) & \xleftarrow{\approx} & \mathrm{Spec}^h(\mathcal{Q}_E^\bullet(K)[T^{-1}]) \end{array}$$

We just proved that $\mathfrak{p}$ belongs to the open subset in the bottom left corner. As K is a proper subgroup of H , we know that β^K is surjective by induction hypothesis and we conclude that $\mathfrak{p}$ belongs to the image of β^H as well. $\square$

8.15. *Remark.* In Theorem 8.13 we have proved something slightly more precise, namely that the map

$$(8.16) \quad \mathcal{Q}_E^\bullet(H) \rightarrow \mathcal{O}_E^\bullet(H) / \langle \xi_\pi^\pm \rangle$$

(where π ranges over all coordinates) is surjective with nilpotent kernel. We expect that $\mathcal{Q}_E^\bullet(H)$ is already reduced, which would imply that (8.16) is in fact an isomorphism of graded rings. In particular, for $p = 2$ we expect that $\mathcal{Q}_E^\bullet(H) \xrightarrow{\sim} \mathcal{O}_E^\bullet(H)$.

9. APPLICATIONS AND EXAMPLES

In this final section, we push our techniques further and compute more examples.

9.1. *Remark.* For an elementary abelian group E , Corollary 6.4 and Theorem 8.13 allow us to think of the *geometry* of $\mathrm{Spc}(\mathcal{K}(E))$, beyond its mere topology, by viewing $\mathrm{Spc}(\mathcal{K}(E))$ as a Dirac scheme. Consider further the ‘periodic’ locus of $\mathrm{Spc}(\mathcal{K}(E))$, which is the open complement of the closed points $\{ \mathcal{M}(H) \mid H \leq E \}$; see Recollection 1.8. This is analogous to considering the projective support variety $\mathrm{Proj}(\mathrm{H}^\bullet(E, k)) \cong \mathbb{P}_k^{r-1}$ by removing the ‘irrelevant ideal’ $\mathcal{M}(1) = \mathrm{H}^+(E, k)$ from $\mathrm{Spec}^h(\mathrm{H}^\bullet(E, k))$. To avoid confusion with the phrase ‘closed points’, we now

refer to the $\mathcal{M}(H)$ as *very closed points*, allowing us to speak of closed points of $\mathbb{P}_k^{r-1}$ in the usual sense (as we did in Corollary 7.12). Removing those finitely many ‘irrelevant’ points allows us to draw more geometric pictures by depicting the (usual) closed points of the periodic locus, as in classical algebraic geometry.

In fact, for any finite group G , we can speak of the *periodic locus* of $\mathrm{Spc}(\mathcal{K}(G))$ to mean the open $\mathrm{Spc}'(\mathcal{K}(G)) := \mathrm{Spc}(\mathcal{K}(G)) \setminus \{ \mathcal{M}(H) \mid H \in \mathrm{Sub}_p(G) \}$ obtained by removing the ‘irrelevant’ very closed points. However, we do not endow these spectra with a scheme-theoretic structure beyond the elementary abelian case, since we do not have Corollary 6.4 in general. We postpone a systematic treatment of the periodic locus to later work. For now we focus on examples.

9.2. Example. Let us revisit Klein-four, with the notation of Example 7.16. From the picture in (7.17) we see that the union of the open subsets $U_E(1)$ and $U_E(E)$ only misses (three) very closed points hence covers the periodic locus. We have

$$(9.3) \quad \begin{aligned} \underline{\mathcal{Q}}_E^\bullet(1) &= \frac{k[\zeta_{N_0}^+, \zeta_{N_1}^+, \zeta_{N_\infty}^+]}{\langle \zeta_{N_0}^+ + \zeta_{N_1}^+ + \zeta_{N_\infty}^+ \rangle} & (= H^*(E; k)), \\ \underline{\mathcal{Q}}_E^\bullet(E) &= \frac{k[\zeta_{N_0}^-, \zeta_{N_1}^-, \zeta_{N_\infty}^-]}{\langle \zeta_{N_0}^- \zeta_{N_1}^- + \zeta_{N_1}^- \zeta_{N_\infty}^- + \zeta_{N_\infty}^- \zeta_{N_0}^- \rangle} \end{aligned}$$

and their homogeneous spectra are both a projective line with a unique closed point added. (For $\underline{\mathcal{Q}}_E^\bullet(E)$, the coordinate transformation for $i = 0, 1$, $\zeta_{N_i}^- \mapsto \tilde{\zeta}_i^- := \zeta_{N_0}^- + \zeta_{N_\infty}^-$, identifies the ring with $k[\tilde{\zeta}_0^-, \tilde{\zeta}_1^-, \zeta_{N_\infty}^-] / \langle \tilde{\zeta}_0^- \tilde{\zeta}_1^- + (\zeta_{N_\infty}^-)^2 \rangle$, which corresponds to the image of a degree-two Veronese embedding of $\mathbb{P}^1$ in $\mathbb{P}^2$.) Removing the very closed points (Remark 9.1), it is a straightforward exercise to check that the two lines are glued along the open complement of the $\mathbb{F}_2$ -rational points, according to the rule $(\zeta_{N_i}^+)^{-1} = \zeta_{N_i}^-$. In other words, we obtain the following picture of $\mathrm{Spc}'(\mathcal{K}(E))$:

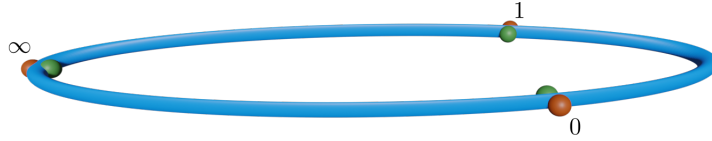


FIGURE 2. A $\mathbb{P}_k^1$ with three doubled points.

To translate between this picture and the one in (7.17), think of the blue part as $\mathbb{P}_k^1$, the three green points as the $\mathbb{F}_2$ -rational points $i = 0, 1, \infty$ in $U_E(1) = V_E$ and the brown points as the $\eta_E(N_i)$ in $U_E(E)$.

9.4. Example. In view of later applications let us consider the action induced on spectra by the involution on $E = C_2 \times C_2$ that interchanges the two C_2 -factors. Let us say that the two factors correspond to the subgroups N_0 and N_1 . On the generators $\zeta_{N_i}^\pm$ of $\underline{\mathcal{Q}}_E^\bullet(1)$ and $\underline{\mathcal{Q}}_E^\bullet(E)$ in (9.3), the effect of the involution is

$$\zeta_{N_0}^\pm \mapsto \zeta_{N_1}^\pm \quad \zeta_{N_1}^\pm \mapsto \zeta_{N_0}^\pm \quad \zeta_{N_\infty}^\pm \mapsto \zeta_{N_\infty}^\pm.$$

The subrings of invariants in $\underline{\mathcal{Q}}_E^\bullet(1)$ and $\underline{\mathcal{Q}}_E^\bullet(E)$ are, respectively,

$$\frac{k[e_1^+, e_2^+, \zeta_{N_\infty}^+]}{\langle e_1^+ + \zeta_{N_\infty}^+ \rangle} \cong k[e_2^+, \zeta_{N_\infty}^+] \quad \text{and} \quad \frac{k[e_1^-, e_2^-, \zeta_{N_\infty}^-]}{\langle e_1^- \zeta_{N_\infty}^- + e_2^- \rangle} \cong k[e_1^-, \zeta_{N_\infty}^-]$$

where $e_1^\pm = \zeta_{N_0}^\pm + \zeta_{N_1}^\pm$ and $e_2^\pm = \zeta_{N_0}^\pm \zeta_{N_1}^\pm$ are the first and second symmetric polynomials in $\zeta_{N_0}^\pm$ and $\zeta_{N_1}^\pm$. Thus $e_i^\pm$ has degree $\pm i$. The homogeneous spectra of these rings (with unique very closed point removed) are again two projective lines⁽²⁾ and they are glued together along the complement of two points. In other words, the quotient of $\mathrm{Spc}'(\mathcal{K}(E))$ by the involution is a $\mathbb{P}_k^1$ with two doubled points:



FIGURE 3. A $\mathbb{P}_k^1$ with two doubled points.

Alternatively, the topological space underlying this quotient may be obtained more directly at the level of Figure 2. Indeed, this involution fixes the two colored points corresponding to ∞ , fixes no other points, and swaps the points corresponding to 0 with the points corresponding to 1, respecting the color. So, again, the quotient can be pictured as a $\mathbb{P}_k^1$ with only two doubled points as in Figure 3.

Let us return to general finite groups. We want to optimize the Colimit Theorem 2.10 by revisiting the category of elementary abelian p -sections $\mathcal{E}_p(G)$.

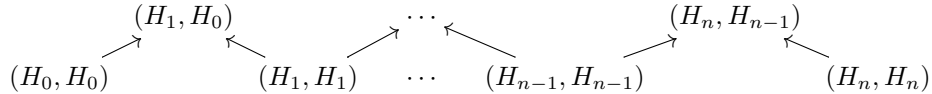
9.5. *Remark.* In Construction 2.1, we gave a ‘raw’ version of the morphisms in the indexing category $\mathcal{E}_p(G)$, which could be fine-tuned without changing the colimit (2.8). As with any colimit, we can quotient-out the indexing category $\mathcal{E}_p(G) \rightarrow \bar{\mathcal{E}}_p(G)$ by identifying any two morphisms that induce the same map by the functor under consideration, here $\mathrm{Spc}(\mathcal{K}(-))$. We then still have

$$(9.6) \quad \mathrm{colim}_{(H,K) \in \mathcal{E}_p(G)} \mathrm{Spc}(\mathcal{K}(H/K)) \xrightarrow{\sim} \mathrm{Spc}(\mathcal{K}(G)).$$

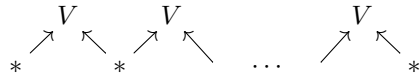
The same holds for any intermediate quotient $\mathcal{E}_p(G) \rightarrow \bar{\mathcal{E}}_p(G) \rightarrow \bar{\bar{\mathcal{E}}}_p(G)$. For instance if $Z(G)$ denotes the center of G , we can consider the category $\bar{\bar{\mathcal{E}}}_p(G)$ obtained from $\mathcal{E}_p(G)$ by modding out the obvious right action of the group $Z(G) \cdot H'$ on each hom set $\mathrm{hom}_{\mathcal{E}_p(G)}((H, K), (H', K'))$.

Let us illustrate how such reductions can be used in practice.

9.7. *Example.* Let $G = C_{p^n}$ be the cyclic group of order p^n . As with any abelian group, using $Z(G) = G$, the reduced category $\bar{\mathcal{E}}_p(G)$ discussed in Remark 9.5 just becomes a poset. Here, if we denote by $1 = H_n < H_{n-1} < \dots < H_1 < H_0 = G$ the tower of subgroups of G then the poset $\bar{\mathcal{E}}_p(G)$ looks as follows:



From Theorem 2.10 we deduce that $\mathrm{Spc}(\mathcal{K}(G))$ is the colimit of the diagram



²More precisely, as already in Example 9.2, we are dealing with weighted projective spaces which happen to be isomorphic to projective lines.

with $*$ = $\mathrm{Spc}(\mathcal{K}(1))$ and $V = \mathrm{Spc}(\mathcal{K}(C_p))$ the V-shaped space in (6.9). In the above diagram, the arrow to the right (resp. left) captures the left-most (resp. right-most) point of V . We conclude that the spectrum of $\mathcal{K}(C_{p^n})$ is equal to

$$(9.8) \quad \begin{array}{ccccccc} m_0 & \bullet & & \bullet & m_1 & & m_{n-1} & \bullet & & \bullet & m_n \\ & \searrow & & \swarrow & & & \swarrow & & \searrow & & \\ p_1 & \bullet & & & & \dots & & & & \bullet & p_n \end{array}$$

This example reproves [BG22, Proposition 7.3]. It will provide the starting point for our upcoming work on the tt-geometry of Artin motives over finite fields.

9.9. Remark. The category of elementary abelian p -sections $\mathcal{E}_p(G)$ is a finite *EI-category*, meaning that all endomorphisms are invertible. The same is true of its reduced versions $\tilde{\mathcal{E}}_p(G)$ and $\bar{\mathcal{E}}_p(G)$ in Remark 9.5. Theorem 2.10 then implies formally that $\mathrm{Spc}(\mathcal{K}(G))$ is the quotient of the spectra for the maximal elementary abelian p -sections by the maximal relations. Let us spell this out.

9.10. Construction. Let I be a finite EI-category. The (isomorphism classes of) objects in I inherit a poset structure with $x \leq y$ if $\mathrm{Hom}_I(x, y) \neq \emptyset$. Maximal objects $\mathrm{Max}(I) \subseteq I$ are by definition the maximal ones in this poset. Now, let x_1, x_2 be two objects in I , possibly equal. The category $\mathrm{Rel}(x_1, x_2)$ of spans $x_1 \leftarrow y \rightarrow x_2$ (or ‘relations’) between x_1 and x_2 , with obvious morphisms (on the y part, compatible with the spans), is also a finite EI-category and we may consider its maximal objects.

9.11. Notation. We denote by $\mathrm{Max}_{p\text{-sec}}^{\mathrm{elab}}(G)$ the set of maximal objects in $\mathcal{E}_p(G)$. A word of warning: In general, there can be more maximal elementary abelian p -sections than just the elementary abelian p -sections of maximal rank.

9.12. Corollary. *Let G be a finite group. The components $\varphi_{(H,K)}$ of (2.7) induce a homeomorphism between the following coequalizer in topological spaces*

$$(9.13) \quad \mathrm{coeq} \left(\coprod_{\substack{E_1 \xleftarrow{g_1} L \xrightarrow{g_2} E_2 \\ \text{maximal relations}}} \mathrm{Spc}(\mathcal{K}(L)) \xrightarrow[\mathrm{Spc}(\mathcal{K}(g_2))]{\mathrm{Spc}(\mathcal{K}(g_1))} \coprod_{E \in \mathrm{Max}_{p\text{-sec}}^{\mathrm{elab}}(G)} \mathrm{Spc}(\mathcal{K}(E)) \right)$$

and $\mathrm{Spc}(\mathcal{K}(G))$, for ‘maximal relations’ in $\mathcal{E}_p(G)$ or any variant of Remark 9.5.

Proof. Applying Theorem 2.10 we obtain

$$\mathrm{Spc}(\mathcal{K}(G)) \simeq \mathrm{coeq} \left(\coprod_{E_1 \xleftarrow{g_1} L \xrightarrow{g_2} E_2} \mathrm{Spc}(\mathcal{K}(L)) \xrightarrow[\mathrm{Spc}(\mathcal{K}(g_2))]{\mathrm{Spc}(\mathcal{K}(g_1))} \coprod_{E \in \mathcal{E}_p(G)} \mathrm{Spc}(\mathcal{K}(E)) \right)$$

where E ranges over *all* elementary abelian p -sections and (g_1, g_2) over *all* relations. There is a canonical map from the coequalizer in the statement to this one and it is straightforward to produce an inverse, as with any finite EI-category. $\square$

We can apply Corollary 9.12 to find the irreducible components of $\mathrm{Spc}(\mathcal{K}(G))$.

9.14. Proposition. *The set of irreducible components of $\mathrm{Spc}(\mathcal{K}(G))$ is in bijection with the set $\mathrm{Max}_{p\text{-sec}}^{\mathrm{elab}}(G)$ of maximal elementary abelian p -sections of G up to conjugation, via the following bijection with generic points:*

$$\begin{aligned} \mathrm{Max}_{p\text{-sec}}^{\mathrm{elab}}(G)/G &\xrightarrow{\sim} \mathrm{Spc}(\mathcal{K}(G))^0 \\ (H, K) &\longmapsto \varphi_{(H,K)}(\eta_{H/K}). \end{aligned}$$

In particular, $\dim(\mathrm{Spc}(\mathcal{K}(G))) = p\text{-rank}_{\mathrm{sec}}(G)$ is the sectional p -rank of G .

Proof. We use coequalizer (9.13). Recall from Proposition 6.11 that $\mathrm{Spc}(\mathcal{K}(E))$ for an elementary abelian p -group E is always irreducible. We get immediately that the map $\mathrm{Max}_{p\text{-sec}}^{\mathrm{elab}}(G)/G \rightarrow \mathrm{Spc}(\mathcal{K}(G))^0$ is a surjection. Assume now that $\varphi_E(\eta_E) = \varphi_{E'}(\eta_{E'})$ for $E, E' \in \mathrm{Max}_{p\text{-sec}}^{\mathrm{elab}}(G)$ and let us show that E and E' are conjugate p -sections. By Corollary 9.12, there exists a finite sequence of maximal relations responsible for the identity $\varphi_E(\eta_E) = \varphi_{E'}(\eta_{E'})$ and we will treat one relation at a time. More precisely, assuming that the generic point in $\mathrm{Spc}(\mathcal{K}(E_1))$ is in the image of (the map on spectra induced by) some relation $E_1 \xleftarrow{g_1} L \xrightarrow{g_2} E_2$, with $E_1, E_2 \in \mathrm{Max}_{p\text{-sec}}^{\mathrm{elab}}(G)$, we will show below that both g_i are conjugation isomorphisms (type (a) in Examples 2.2). In particular, E_1, E_2 are conjugate. And as conjugation identifies the unique generic points in the spectra for E_1 and E_2 one can apply induction on the number of relations to conclude.

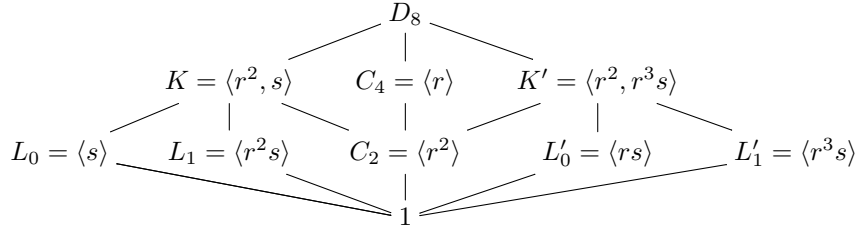
As the map induced by g_1 is a closed immersion (Lemma 2.9) it must be a homeomorphism once its image contains the generic point. From this, we deduce that g_1 itself must be an isomorphism. (Indeed, the map induced by restriction to a proper subgroup of E_1 is not surjective, already on the cohomological open. And similarly, the map induced by modular fixed-points with respect to a non-trivial subgroup of E_1 does not even meet the cohomological open.) Hence $L \simeq E_1$ is maximal too and therefore g_2 is also an isomorphism. The only isomorphisms in $\mathcal{E}_p(G)$ are conjugations (Remark 2.3) and we conclude.

The second statement follows from this together with Proposition 6.14. $\square$

9.15. *Remark.* For G not elementary abelian, we already saw with the example of $G = Q_8$ in [BG22, Example 7.12] that $\mathrm{Spc}(\mathcal{K}(G))$ can have larger Krull dimension than $\mathrm{Spc}(\mathrm{D}_b(kG))$. And indeed, Q_8 has sectional p -rank two and p -rank one.

9.16. *Remark.* For every maximal $(H, K) \in \mathrm{Max}_{p\text{-sec}}^{\mathrm{elab}}(G)$, since $\varphi_{(H,K)}$ is a closed map, it yields a surjection $\varphi_{(H,K)}: \mathrm{Spc}(\mathcal{K}(E)) \rightarrow \overline{\{\varphi_{H,K}(\eta_{H/K})\}}$ from the spectrum of the elementary abelian $E = H/K$ onto the corresponding irreducible component of $\mathrm{Spc}(\mathcal{K}(G))$. We illustrate this with $G = D_8$ in Example 9.17 below, where said surjection coincides with the folding of Example 9.4.

9.17. *Example.* We will now explain the meaning of Figure 1 and in effect compute $\mathrm{Spc}(\mathcal{K}(D_8))$ for $G = D_8 = \langle r, s \mid r^4 = s^2 = 1, rs = sr^{-1} \rangle$ the dihedral group of order 8. We label its subgroups as follows⁽³⁾:



Since L_0 and L_1 (resp. L'_0 and L'_1) are G -conjugate, by the element r , we have exactly eight very closed points $\mathcal{M}(H)$ for $H \in \mathrm{Sub}_p(G)/G$. We shall focus on the open complement of these very closed points, i.e. the periodic locus $\mathrm{Spc}'(\mathcal{K}(D_8))$

³The two Klein-four subgroups are called K and K' . The names L_0 and L_1 for the cyclic subgroups of K (resp. L'_0 and L'_1 in K') are chosen to evoke N_0 and N_1 in Example 7.16. The third cyclic subgroup, N_∞ , corresponds to $C_2 = Z(D_8)$ and is common to K and K' .

of Remark 9.1, which is of Krull dimension one. Since all maps in the coequalizer diagram (9.13) preserve the dimension of points (Lemma 2.9) we may first remove these very closed points and *then* compute the coequalizer.

Let us describe $\text{Max}_{p\text{-sec}}^{\text{elab}}(D_8)$ and the maximal relations. In addition to the maximal elementary abelian subgroups K and K' there is one maximal elementary abelian subquotient D_8/C_2 . So we have three maximal sections: $\text{Max}_{p\text{-sec}}^{\text{elab}}(D_8) = \{(K, 1), (K', 1), (D_8, C_2)\}$. We compute the relations in the category $\tilde{\mathcal{E}}_2(D_8)$ which is obtained from $\mathcal{E}_2(D_8)$ by quotienting each hom-set $\text{hom}((H, M), (H', M'))$ by the action of H' , as in Remark 9.5. One then easily finds by inspection five non-degenerate⁽⁴⁾ maximal relations up to isomorphism, pictured as follows:

$$(9.18) \quad \begin{array}{ccccc} & & (D_8, C_2) & & \\ & \nearrow & & \nwarrow & \\ (K, C_2) & & & & (K', C_2) \\ \nearrow & & & & \nwarrow \\ (K, 1) & \xleftarrow{\quad} & (C_2, 1) & \xrightarrow{\quad} & (K', 1) \\ \nwarrow & & & & \nearrow \\ & & & & \end{array} \quad \begin{array}{c} r \\ r \end{array}$$

Here, the loops labeled r represent the relations $(K, 1) \xleftarrow{1} (K, 1) \xrightarrow{r} (K, 1)$, and similarly for K' . All unlabeled arrows are given by $1 \in D_8$, as in Examples 2.2 (b)-(c). We explain below the brown/green color-coding in the other three relations.

Hence the space $\text{Spc}'(\mathcal{K}(D_8))$ is a quotient of three copies of the space $\text{Spc}'(\mathcal{K}(E))$ for E the Klein-four group, equal to $\mathbb{P}_k^1$ with three doubled points as in Figure 2.

Let us discuss the relations. We start with the self-relation corresponding to the loop r on $(K, 1)$. As the conjugation by r on K simply swaps the subgroups L_0 and L_1 , we deduce from Example 9.4 that the quotient of $\text{Spc}'(\mathcal{K}(K))$ by this relation is a $\mathbb{P}_k^1$ with two doubled points, as in Figure 3. The same is true for K' .

At this stage we have identified the three irreducible components (see Figure 4) and the three remaining relations will tell us how to glue these components.

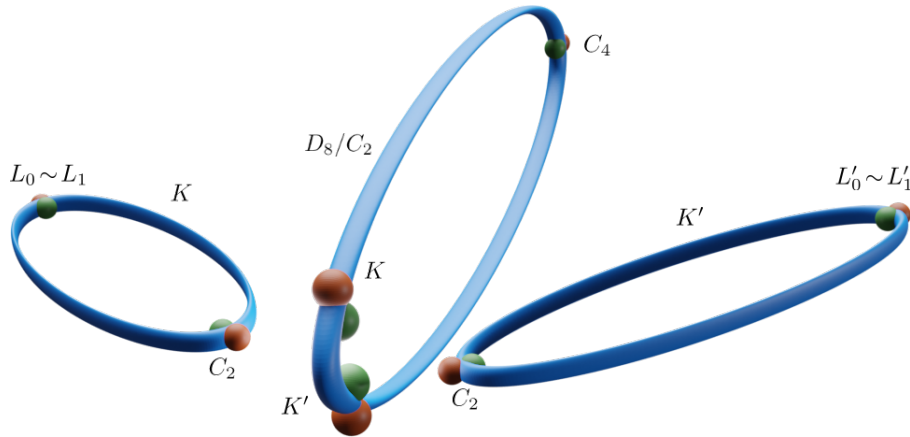


FIGURE 4. Three $\mathbb{P}_k^1$ with several points doubled.

The three sides of the ‘triangle’ (9.18) display maximal relations that identify a single point of one irreducible component with a single point of another. Indeed,

⁴ that is, not of the form $x \xleftarrow{\text{id}} x \xrightarrow{\text{id}} x$ (which would not affect the coequalizer (9.13) anyway)

each of the middle sections K/C_2 , K'/C_2 and $C_2/1$ is a C_2 , whose periodic locus is a single point η_{C_2} (Example 6.8). Each edge in (9.18) identifies the image of that single point η_{C_2} in the two corresponding irreducible components in Figure 4. The color in (9.18) records the color of that image: A brown point or a green point in the $\mathbb{P}_k^1$ with doubled points. Let us do all three. First, the relation between the two Klein-fours, K and K' , at the bottom of (9.18), identifies the two green points corresponding to C_2 , as we are used to with projective support varieties. Then, the last two relations in (9.18), on the sides, identify a brown point in the K - or K' -component with the green point in the D_8/C_2 -component corresponding to K/C_2 and K'/C_2 , respectively. This is a direct verification, for instance using that $(\psi^{C_2})^{-1}(\text{Im}(\rho_K^{D_8})) = (\psi^{C_2})^{-1}(\text{supp}_{D_8}(k(D_8/K))) = \text{supp}_{D_8/C_2}(\Psi^{C_2}(k(D_8/K))) = \text{supp}_{D_8/C_2}(k(D_8/K)) = \text{Im}(\rho_{K/C_2}^{D_8/C_2})$ in $\text{Spc}(\mathcal{K}(D_8/C_2))$.

Thus we obtain $\text{Spc}'(\mathcal{K}(D_8))$ from these three identifications on the space of Figure 4. The result was depicted in Figure 1.

REFERENCES

- [Bal10] Paul Balmer. Spectra, spectra, spectra - tensor triangular spectra versus Zariski spectra of endomorphism rings. *Algebraic and Geometric Topology*, 10(3):1521–63, 2010.
- [Bal16] Paul Balmer. Separable extensions in tensor-triangular geometry and generalized Quillen stratification. *Ann. Sci. Éc. Norm. Supér. (4)*, 49(4):907–925, 2016.
- [BCH⁺23] T. Barthel, N. Castellana, D. Heard, N. Naumann, and L. Pol. Quillen stratification in equivariant homotopy theory, 2023. Preprint: <https://arxiv.org/abs/2301.02212>.
- [BCR97] D. J. Benson, Jon F. Carlson, and Jeremy Rickard. Thick subcategories of the stable module category. *Fund. Math.*, 153(1):59–80, 1997.
- [BG22] Paul Balmer and Martin Gallauer. The tt-geometry of permutation modules. Part I: Stratification, 2022. Preprint: <https://arxiv.org/abs/2210.08311>.
- [BG23] Paul Balmer and Martin Gallauer. Finite permutation resolutions. *Duke Math. J.*, 172(2):201–229, 2023.
- [BIK11] David J. Benson, Srikanth B. Iyengar, and Henning Krause. Stratifying modular representations of finite groups. *Ann. of Math. (2)*, 174(3):1643–1684, 2011.
- [BT08] Serge Bouc and Jacques Thévenaz. Gluing torsion endo-permutation modules. *J. Lond. Math. Soc. (2)*, 78(2):477–501, 2008.
- [Car96] Jon F. Carlson. *Modules and group algebras*. Lectures in Mathematics ETH Zürich. Birkhäuser Verlag, Basel, 1996. Notes by Ruedi Suter.
- [DHM24] Daniel Dugger, Christy Hazel, and Clover May. Equivariant $\mathbb{Z}/\ell$ -modules for the cyclic group C_2 . *Journal of Pure and Applied Algebra*, 228(3):107473, 2024.
- [HP23] Lars Hesselholt and Piotr Pstrągowski. Dirac geometry I: Commutative algebra. *Peking Math. J.*, 2023. To appear: <https://doi.org/10.1007/s42543-023-00072-6>.
- [Lau23] Eike Lau. The Balmer spectrum of certain Deligne–Mumford stacks. *Compositio Mathematica*, 159(6):1314–1346, 2023.
- [Qui71] Daniel Quillen. The spectrum of an equivariant cohomology ring. I, II. *Ann. of Math. (2)*, 94:549–572; *ibid. (2)* **94** (1971), 573–602, 1971.
- [Ser65] Jean-Pierre Serre. Sur la dimension cohomologique des groupes profinis. *Topology*, 3:413–420, 1965.

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